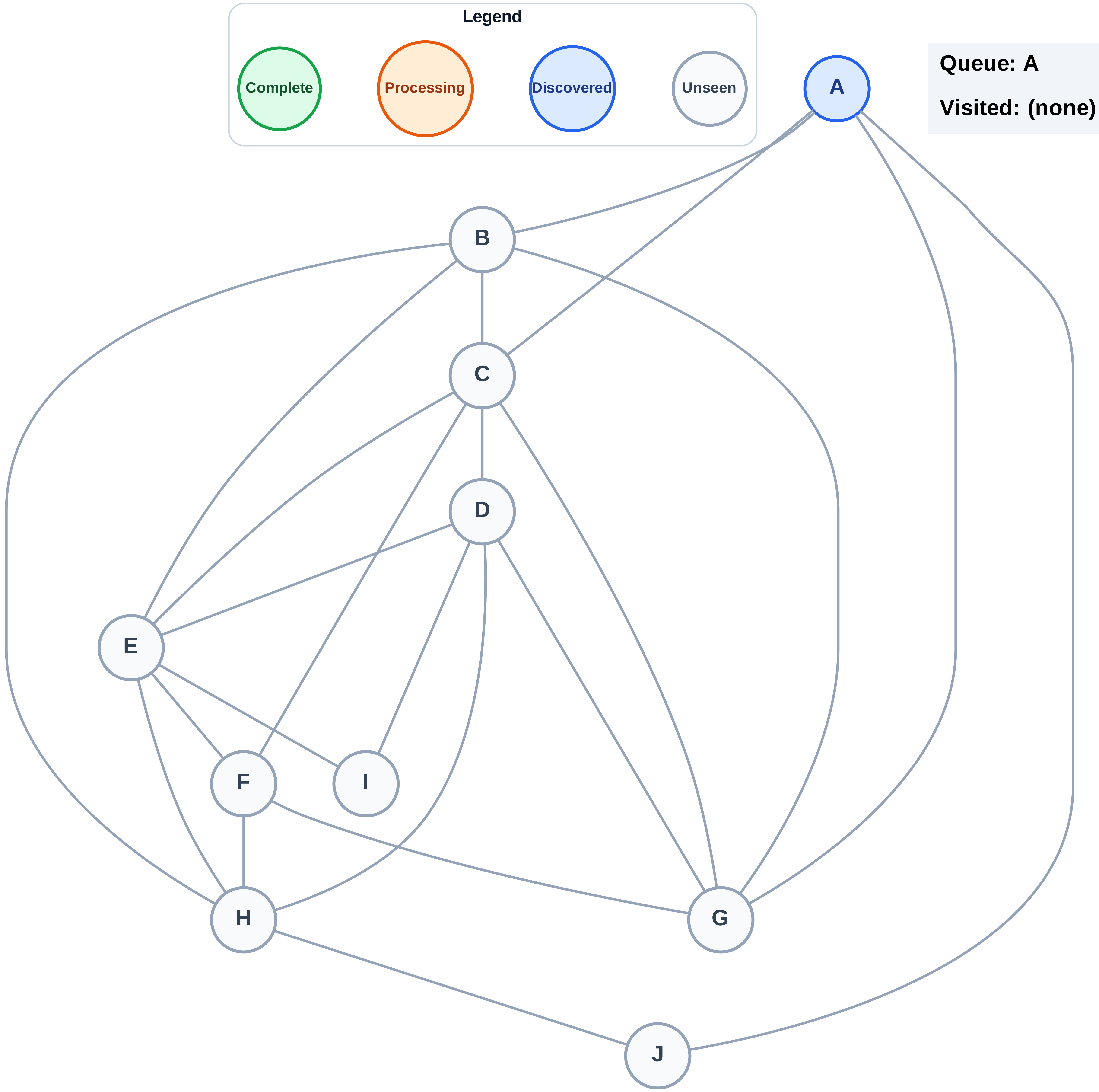
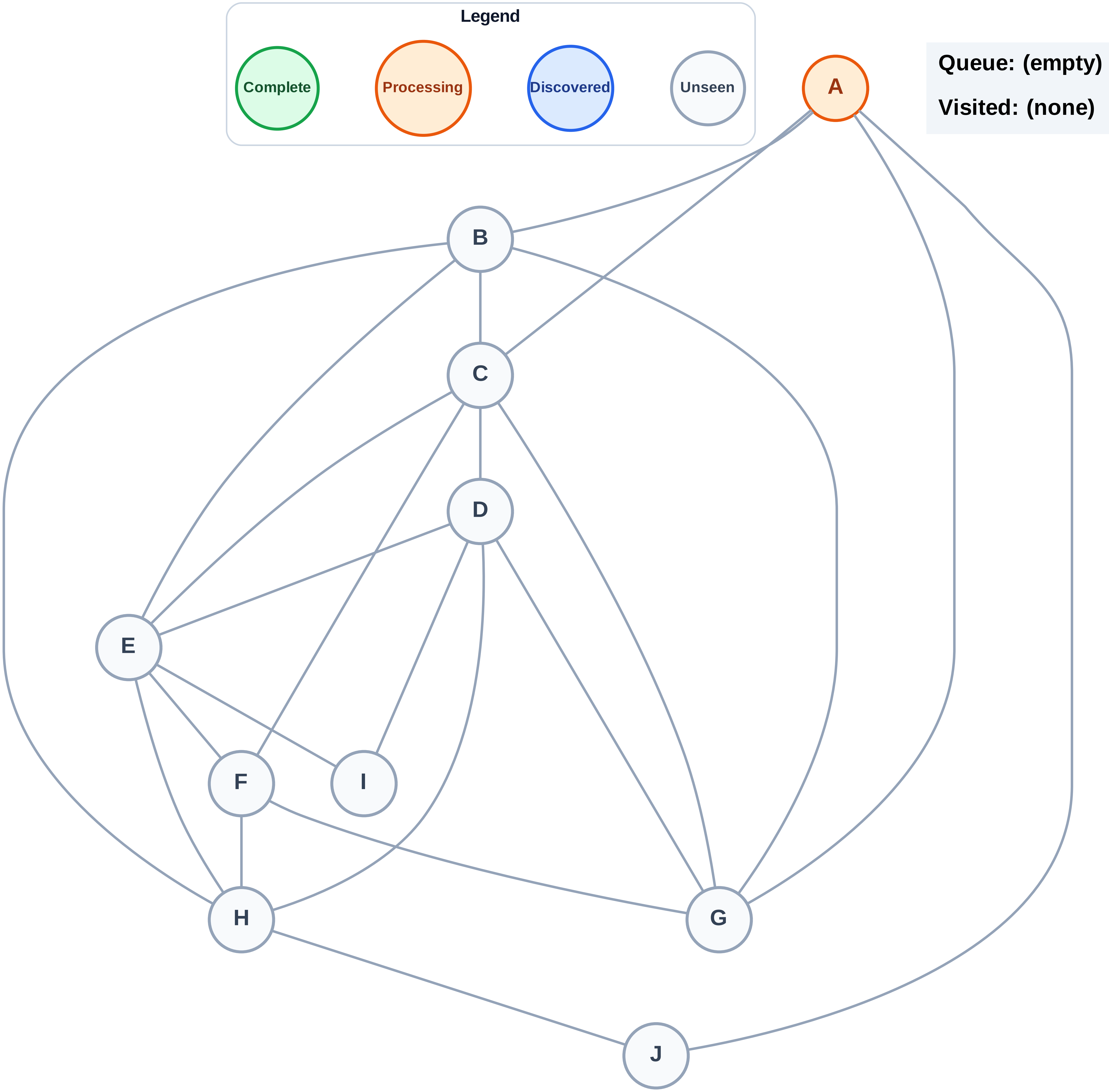
BFS Traversal

Step 1: Start at A



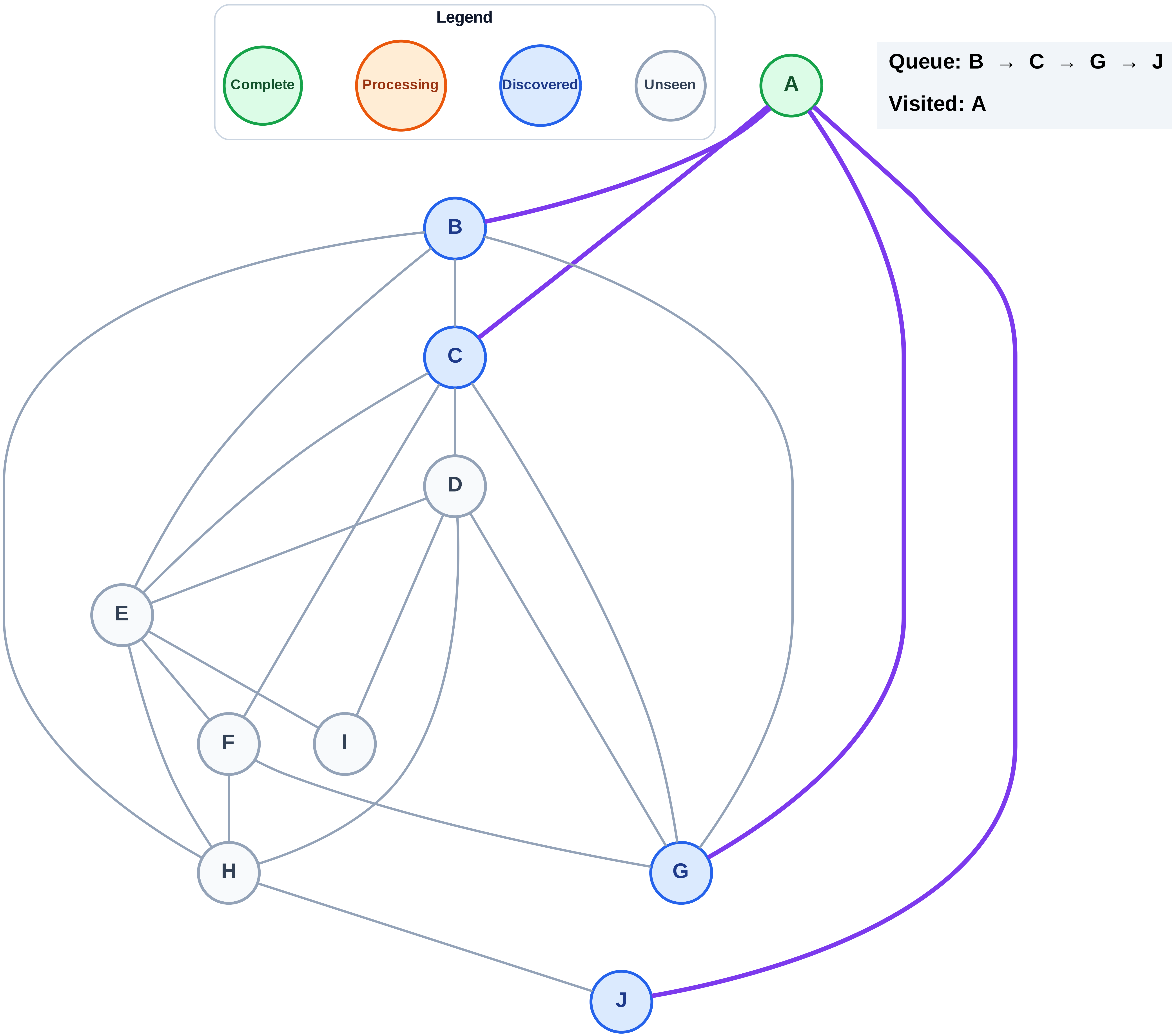
BFS Traversal

Step 2: Process node A



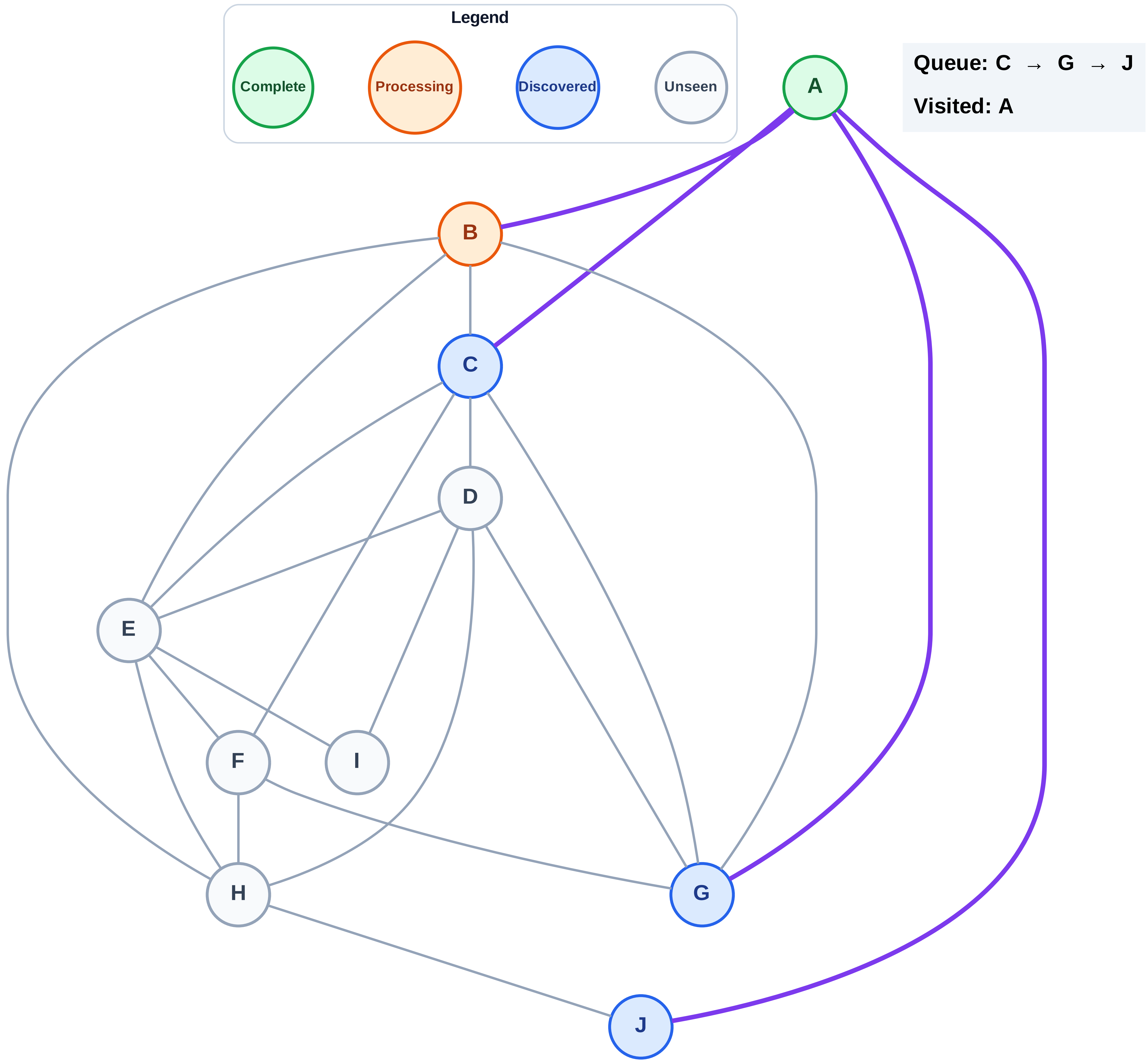
BFS Traversal

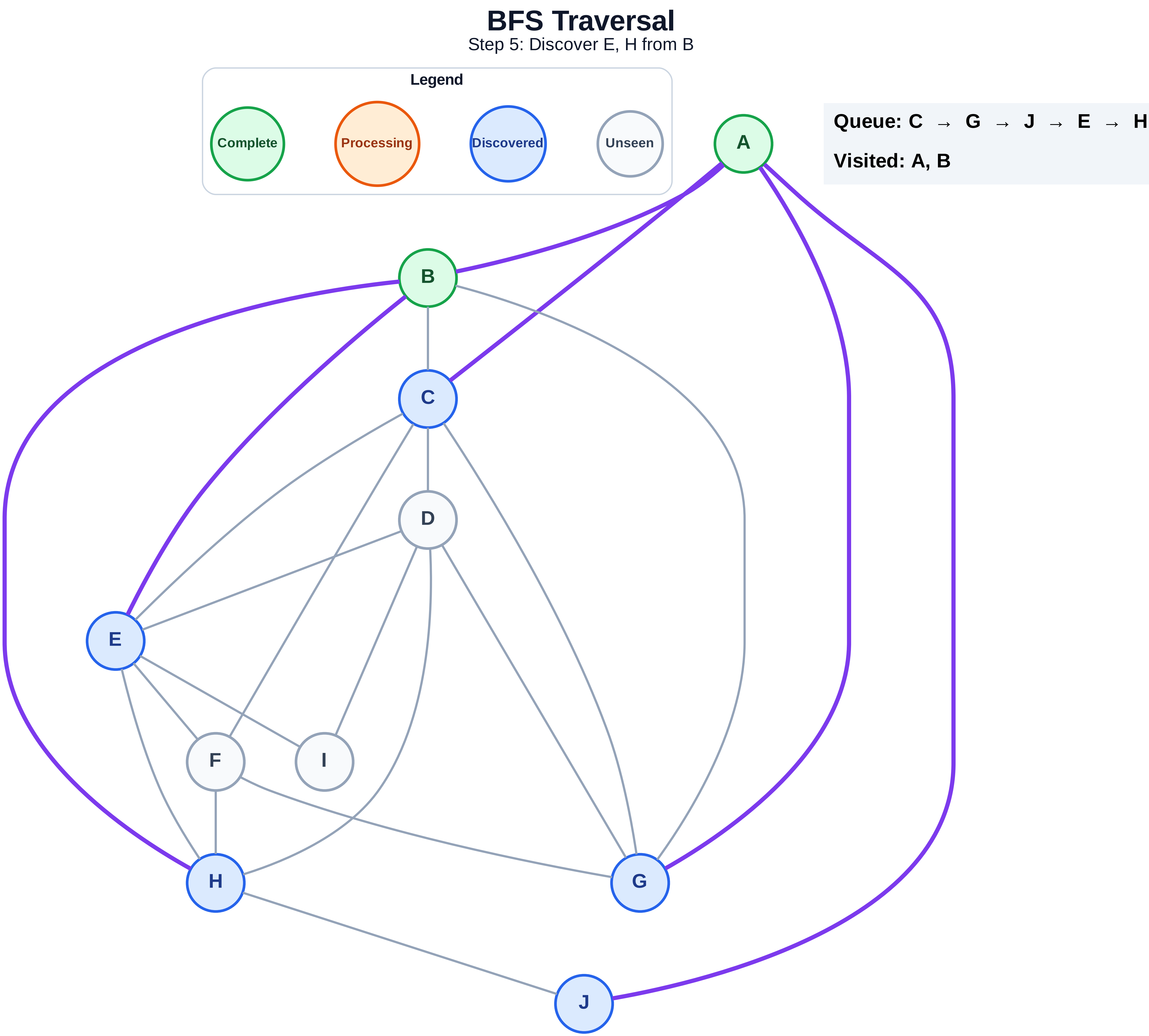
Step 3: Discover B, C, G, J from A



BFS Traversal

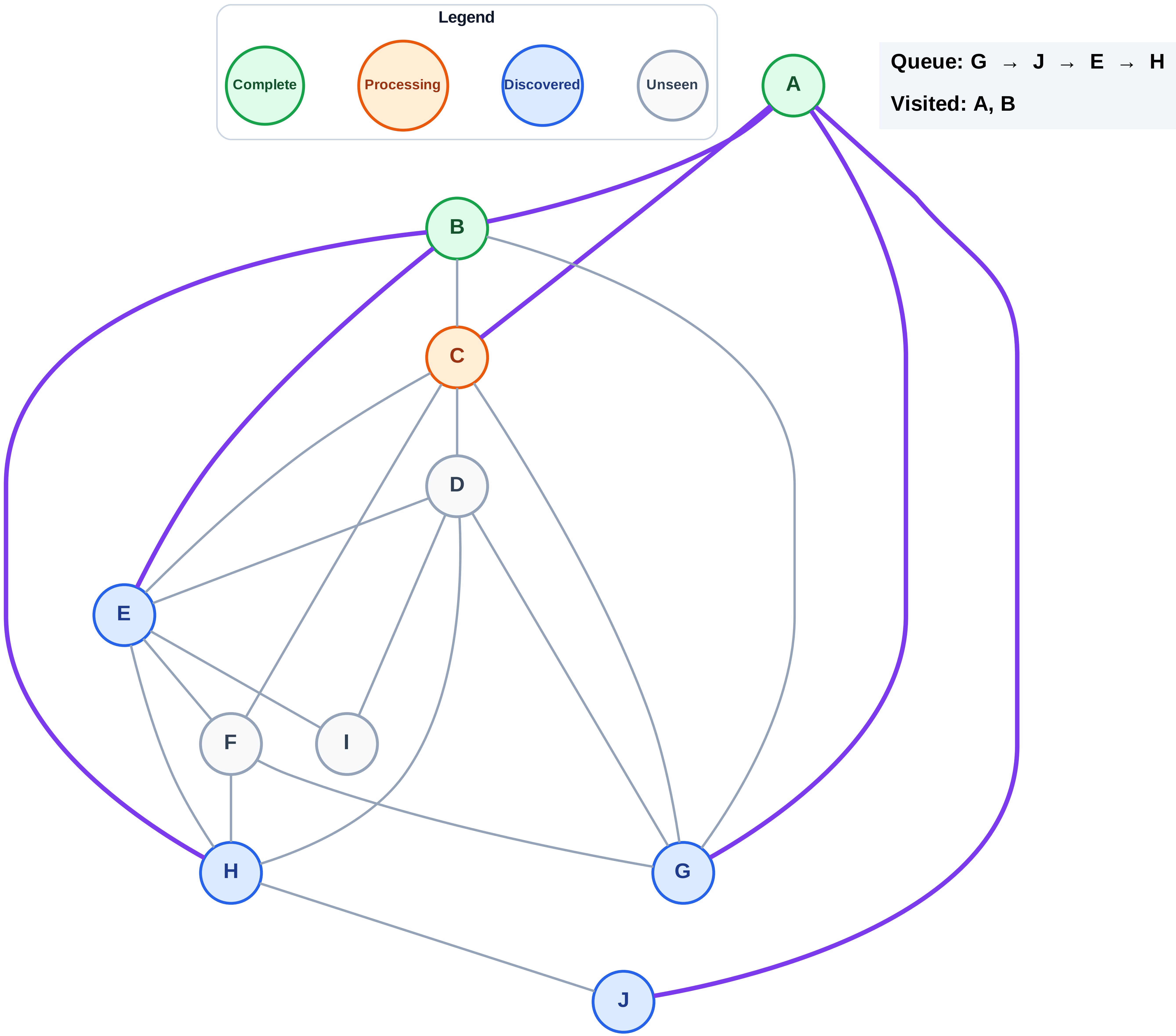
Step 4: Process node B




BFS Traversal
Step 5: Discover E, H from B
Legend
Complete
Processing
Discovered
Unseen
Queue: C → G → J → E → H
Visited: A, B
A
B
C
D
E
F
I
H
G
J

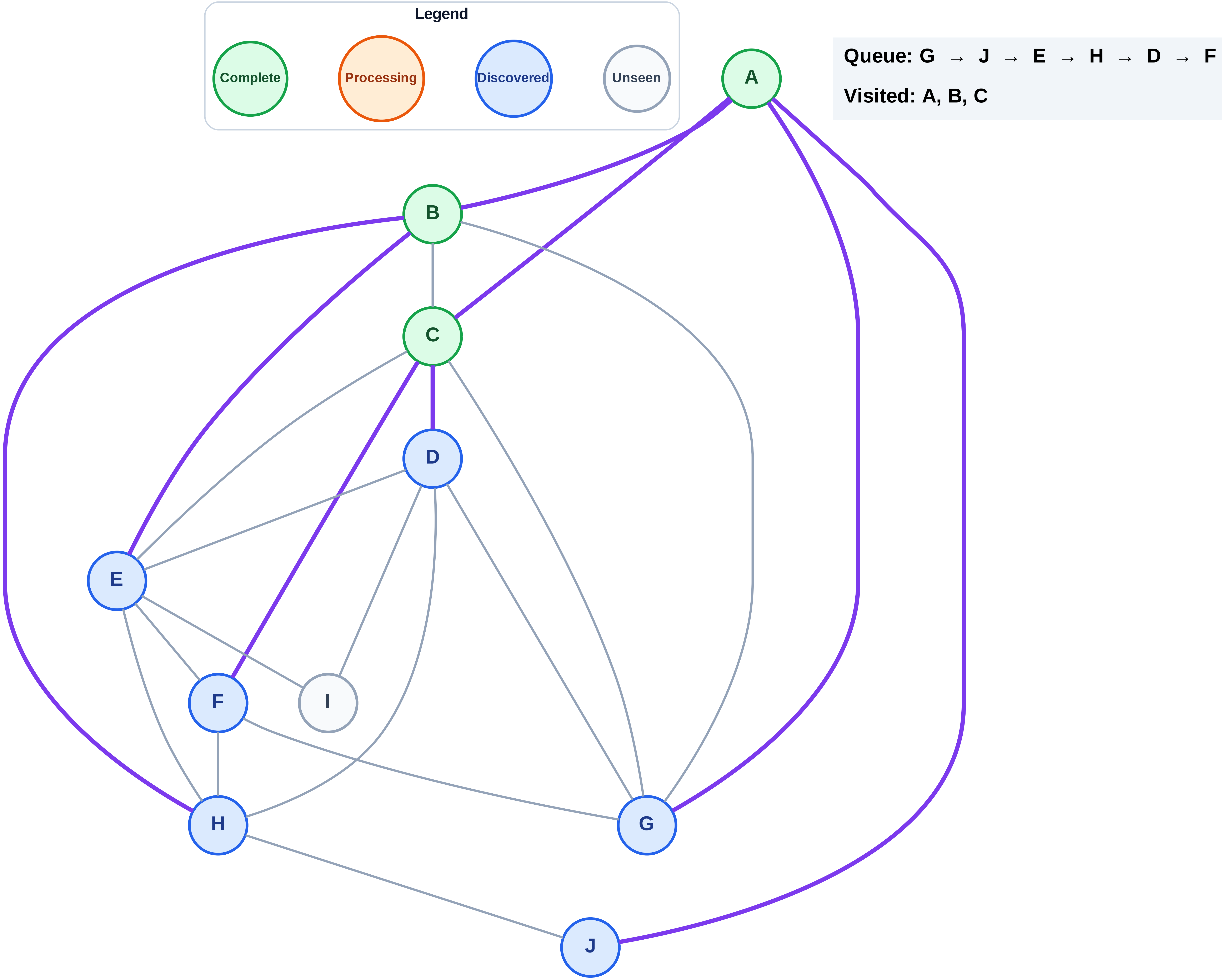
BFS Traversal

Step 6: Process node C



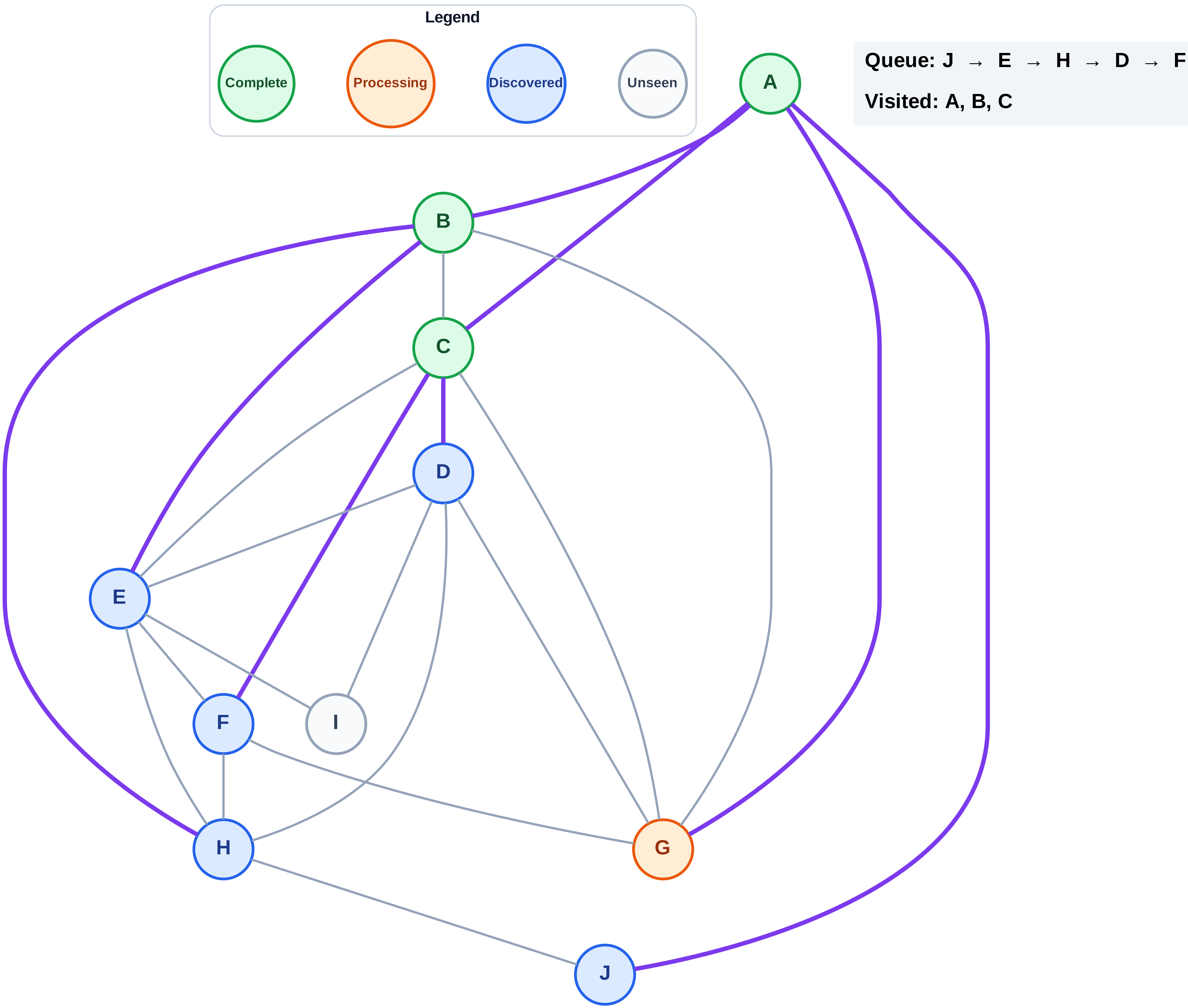
BFS Traversal

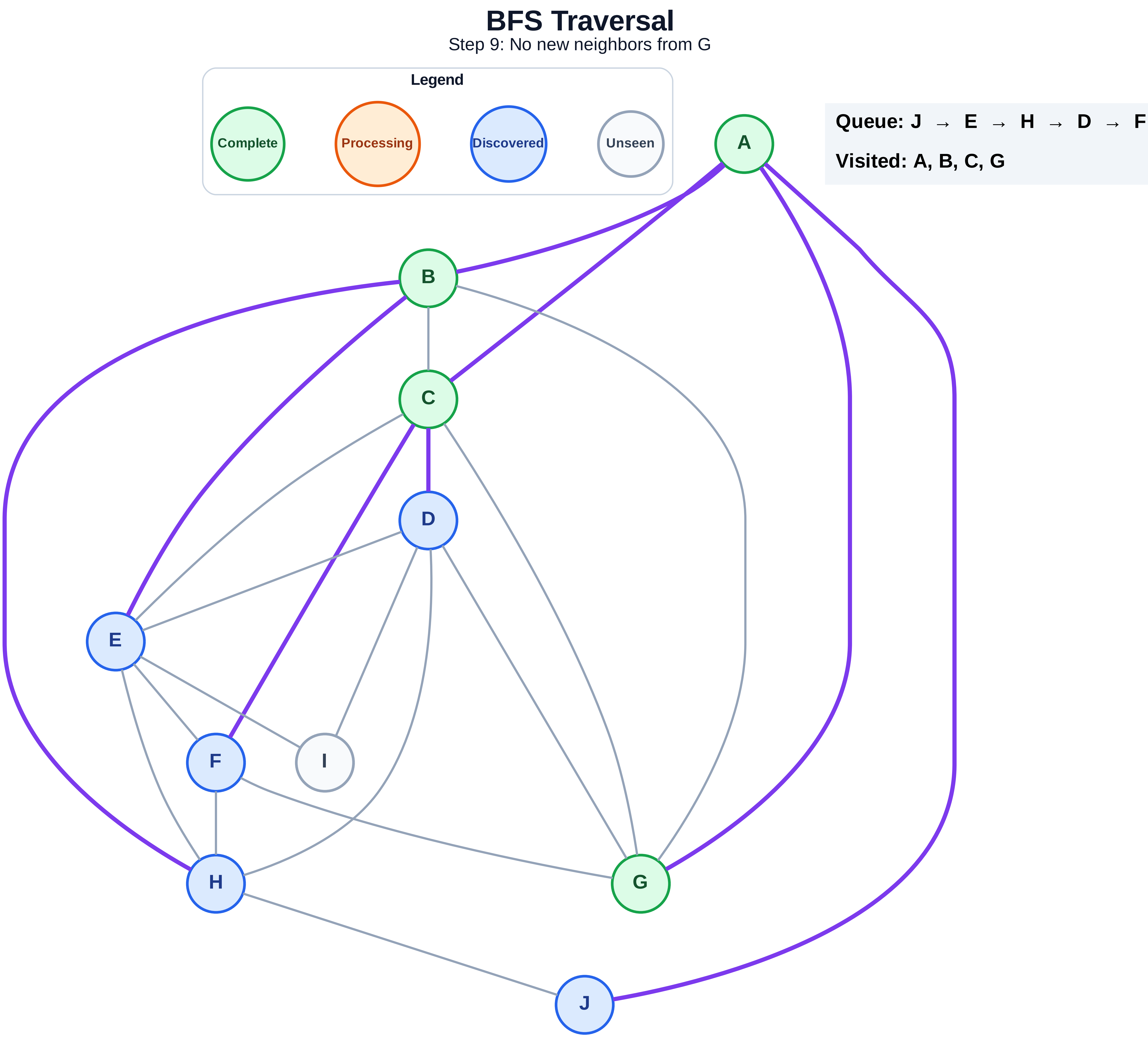
Step 7: Discover D, F from C

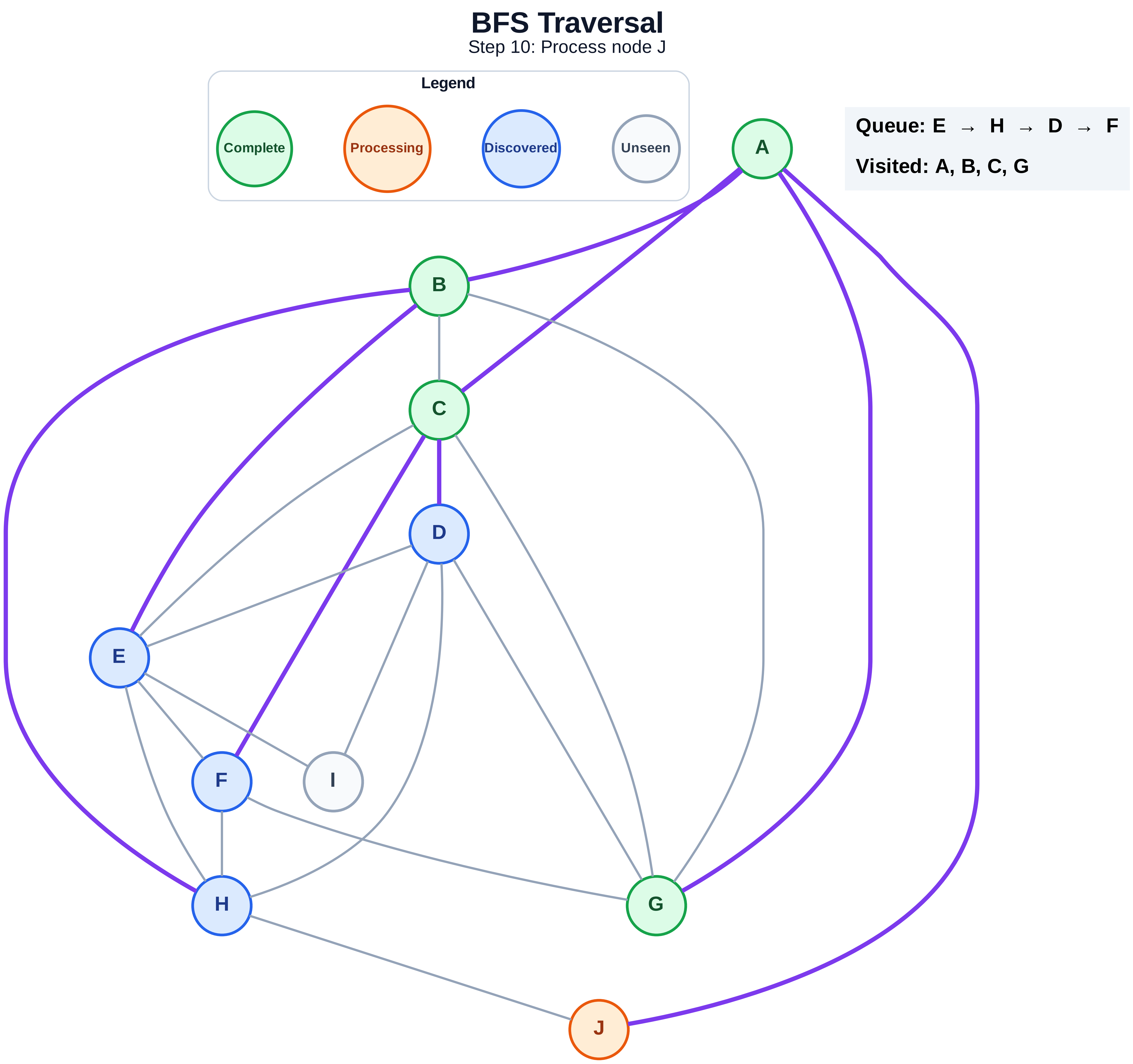


BFS Traversal

Step 8: Process node G

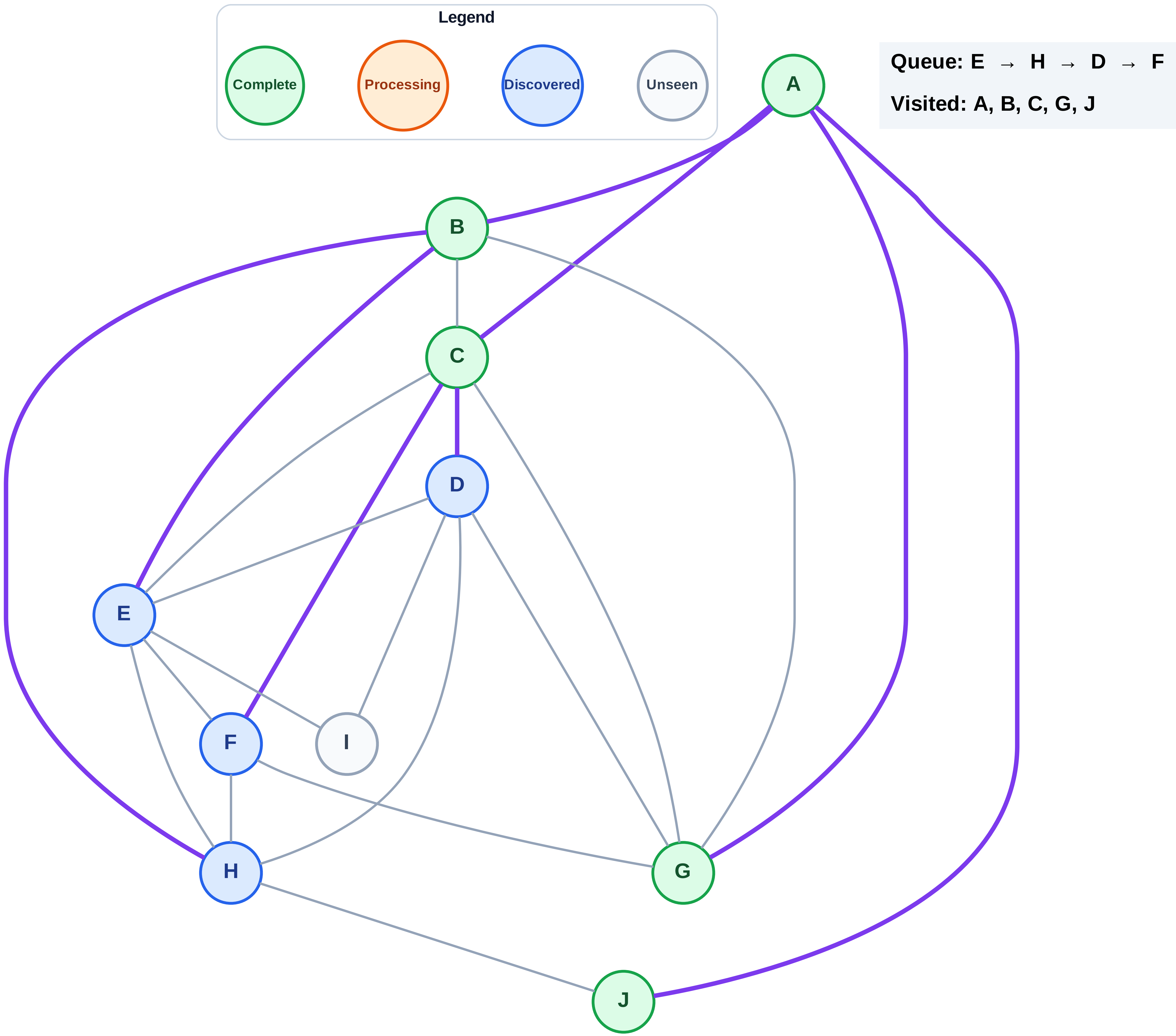



BFS Traversal
Step 9: No new neighbors from G
Legend
Complete
Processing
Discovered
Unseen
Queue: J → E → H → D → F
Visited: A, B, C, G
A
B
C
D
E
F
I
H
G
J


BFS Traversal
Step 10: Process node J
Legend
Complete
Processing
Discovered
Unseen
Queue: E → H → D → F
Visited: A, B, C, G
A
B
C
D
E
F
I
H
G
J

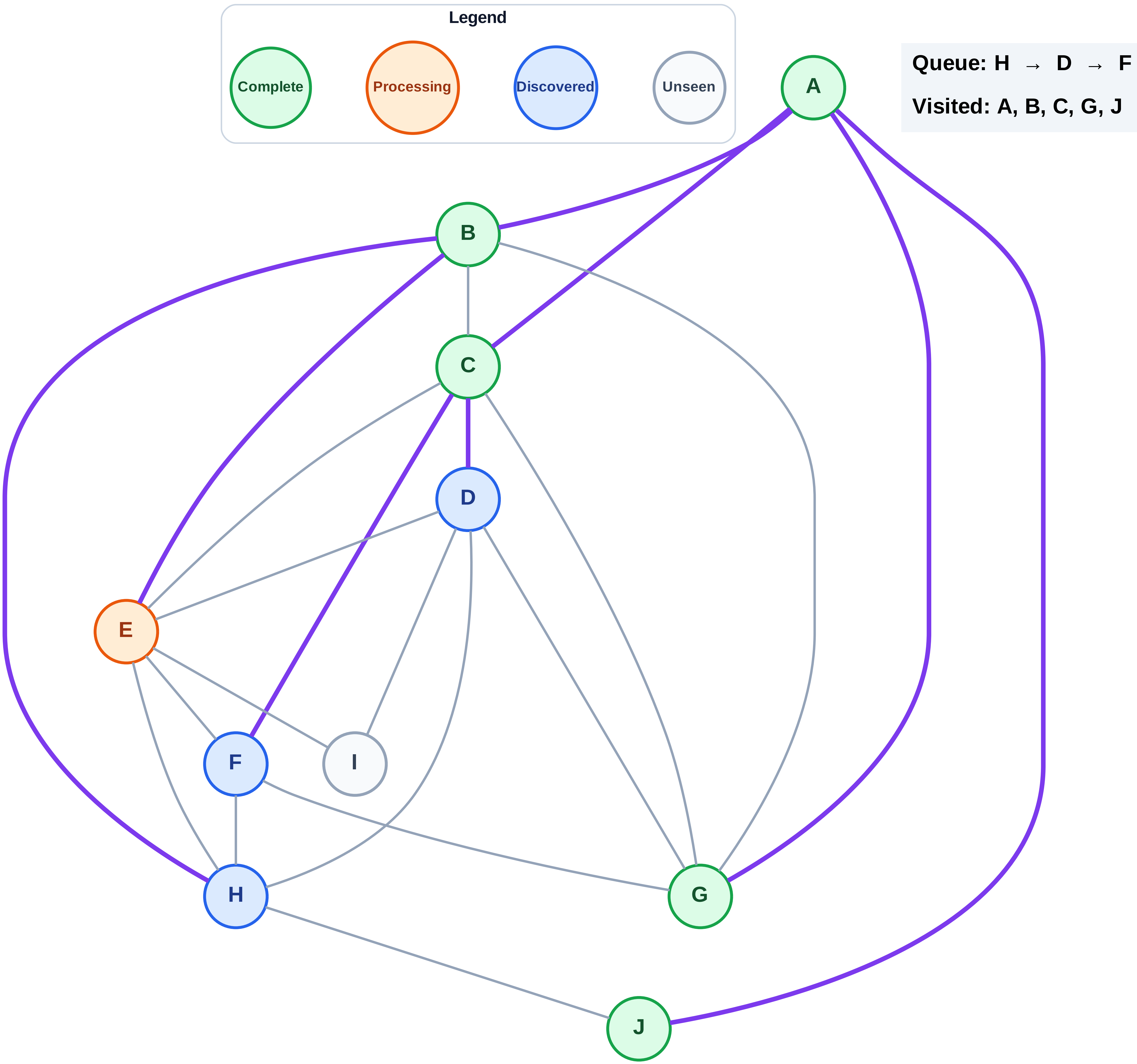
BFS Traversal

Step 11: No new neighbors from J



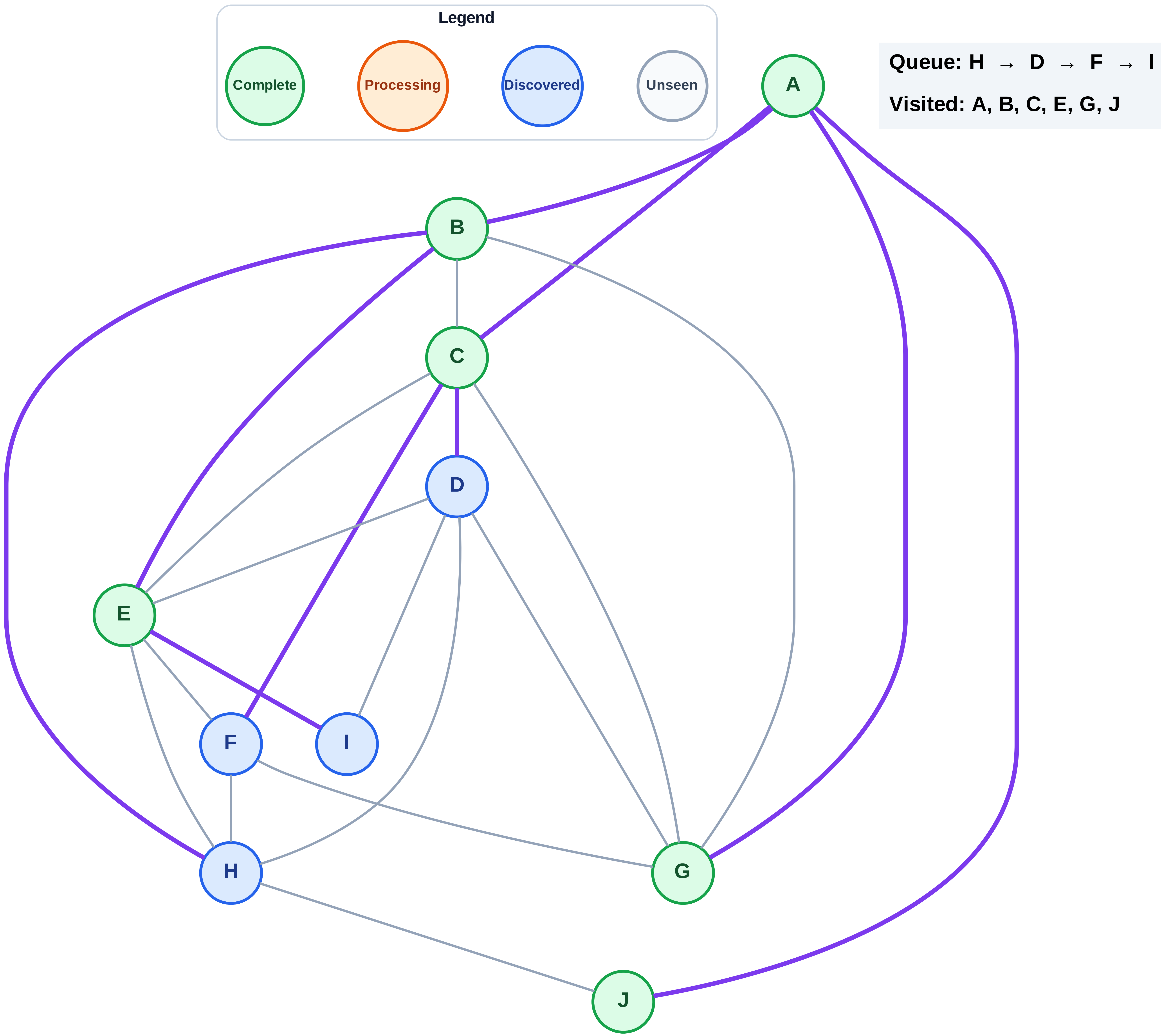
BFS Traversal

Step 12: Process node E



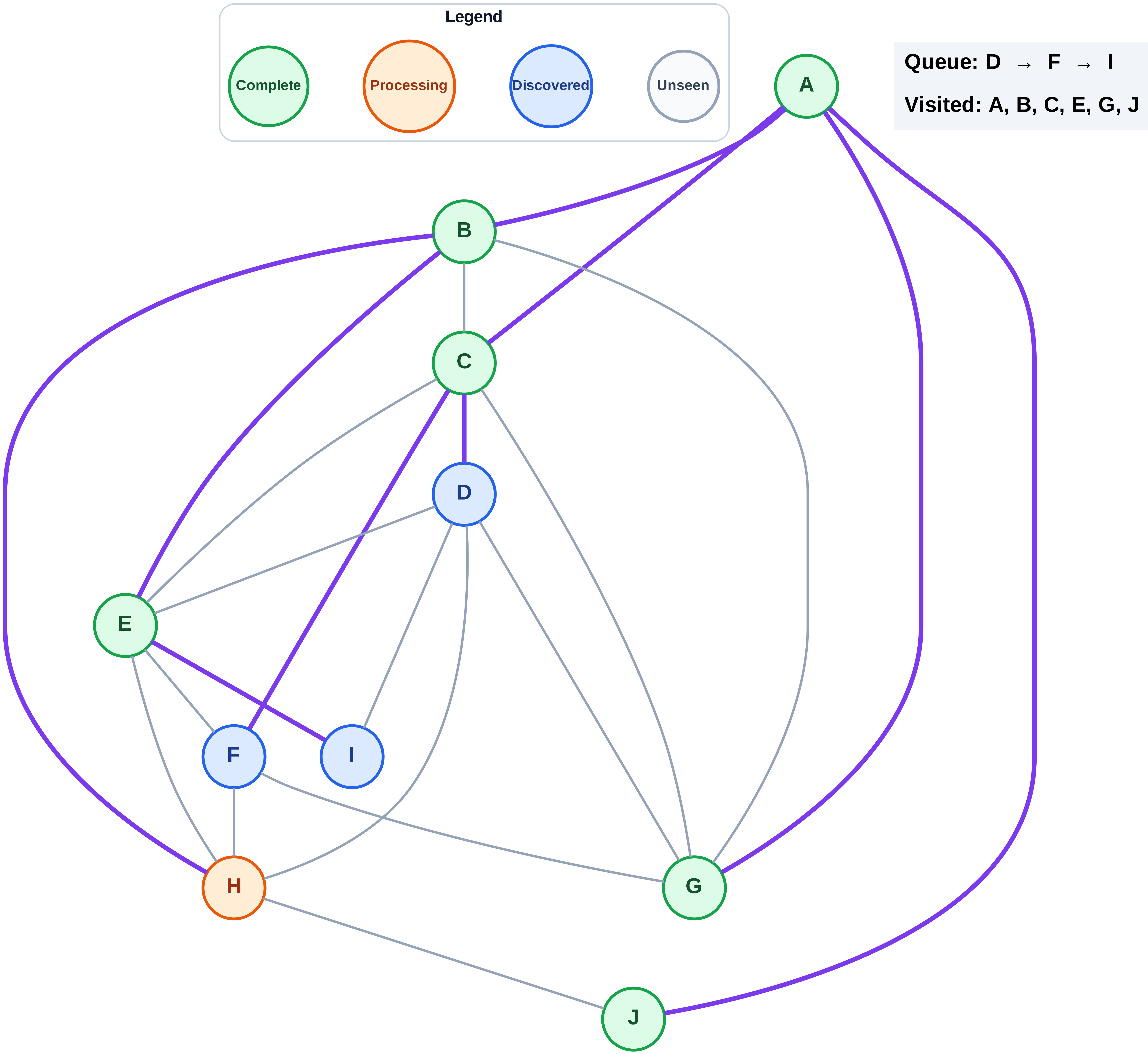
BFS Traversal

Step 13: Discover I from E



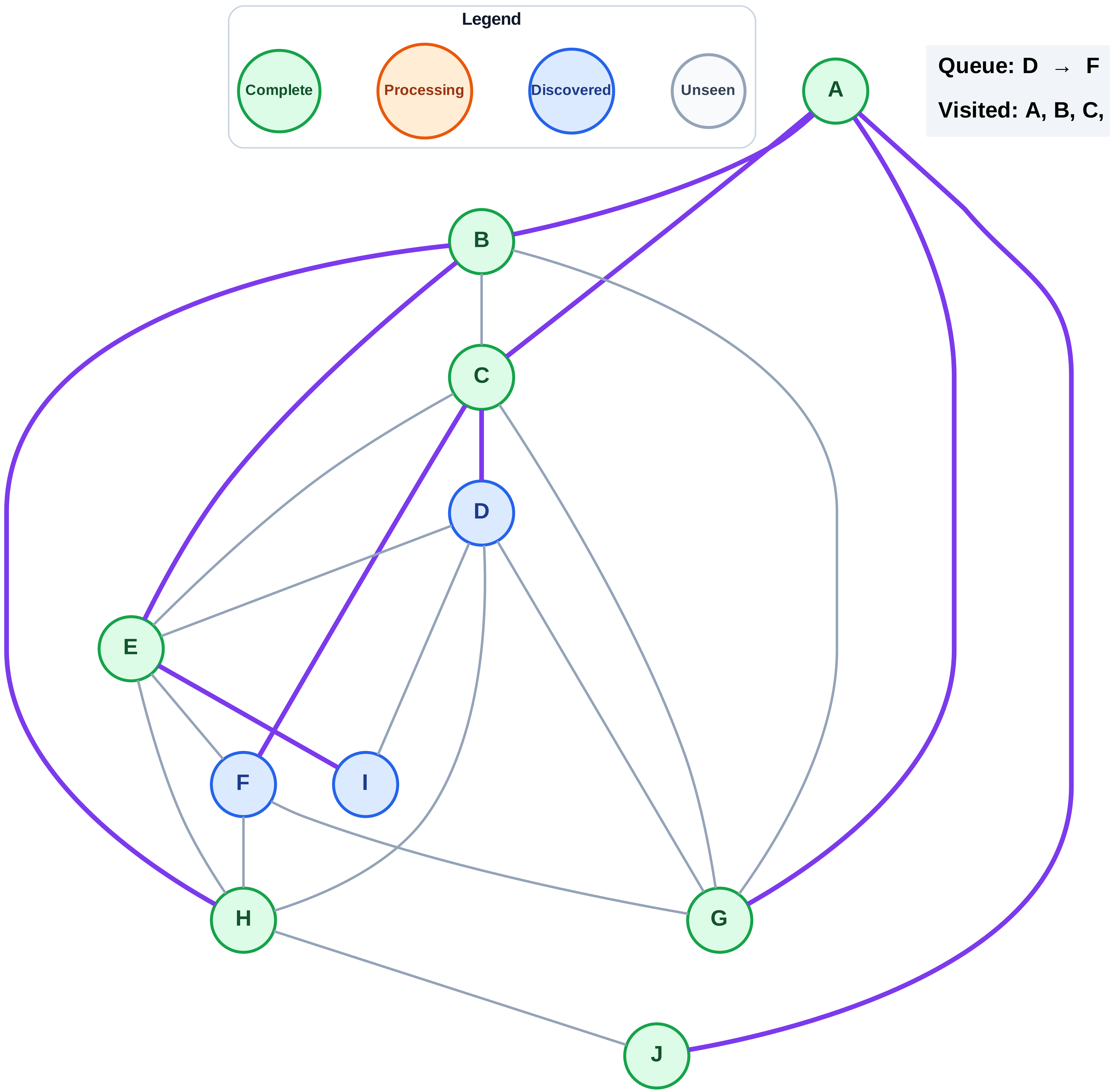
BFS Traversal

Step 14: Process node H



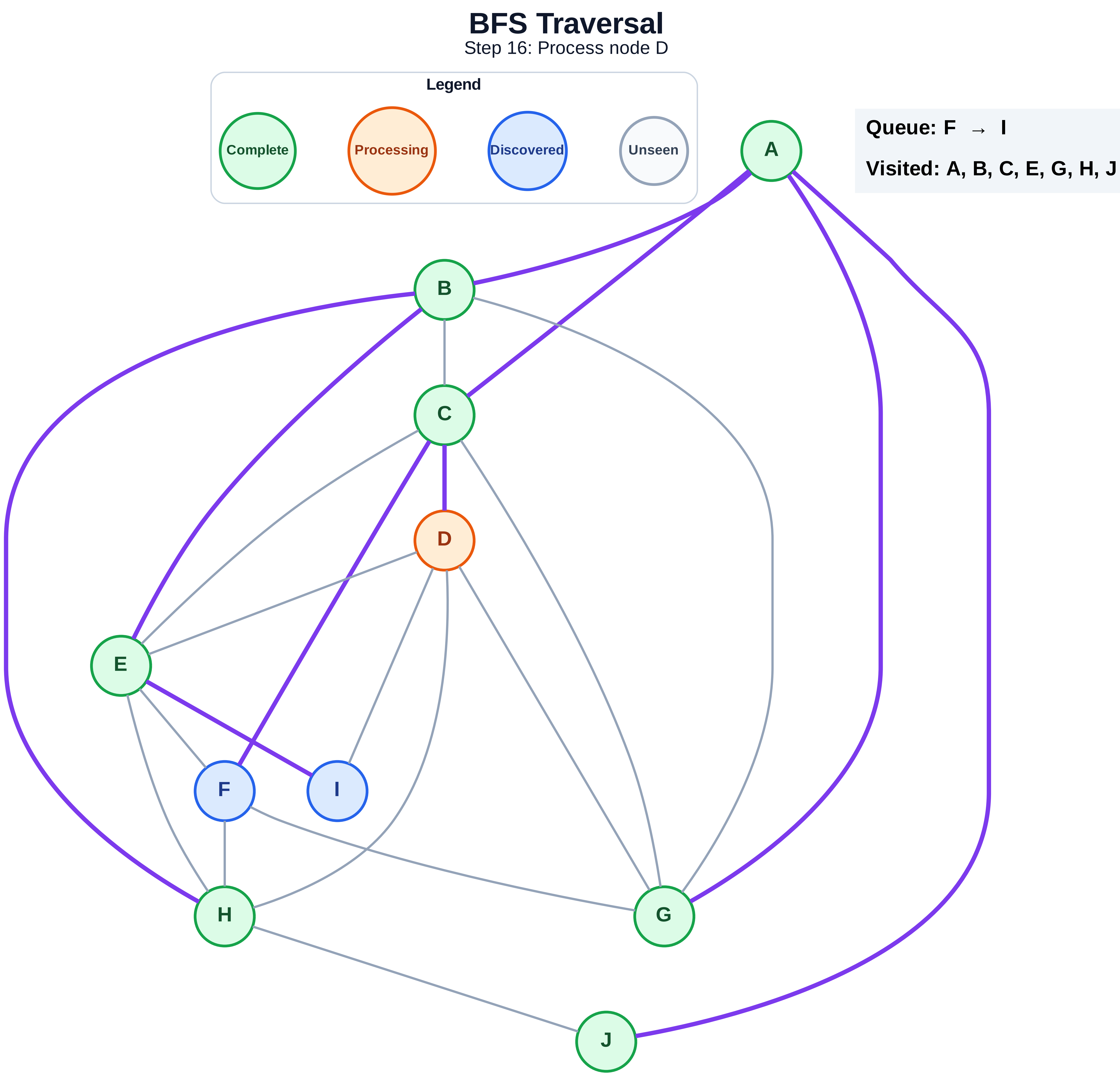
BFS Traversal

Step 15: No new neighbors from H



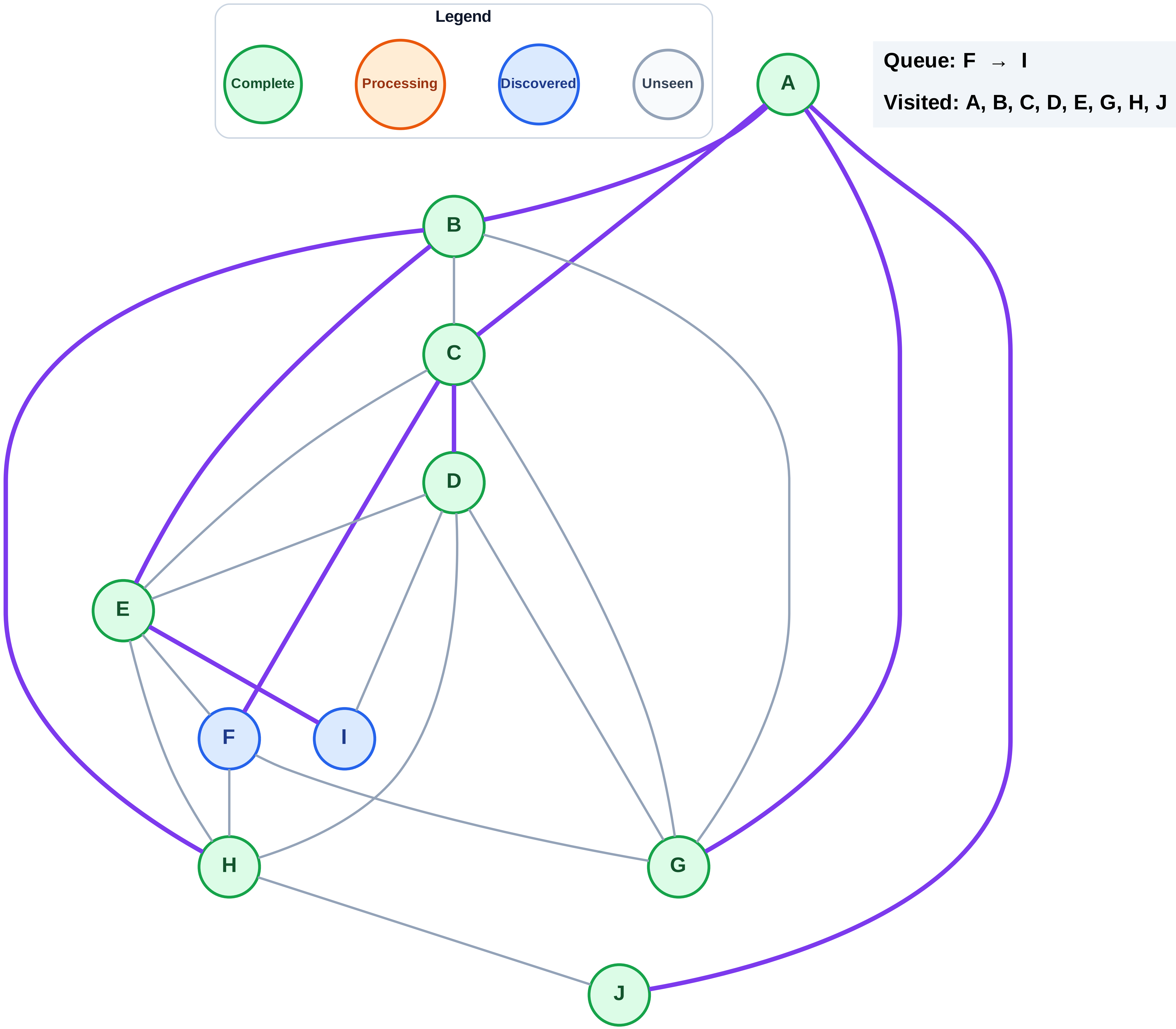
Queue: D → F → I

Visited: A, B, C, E, G, H, J


BFS Traversal
Step 16: Process node D
Legend
Complete
Processing
Discovered
Unseen
Queue: F → I
Visited: A, B, C, E, G, H, J
A
B
C
D
E
F
I
H
G
J

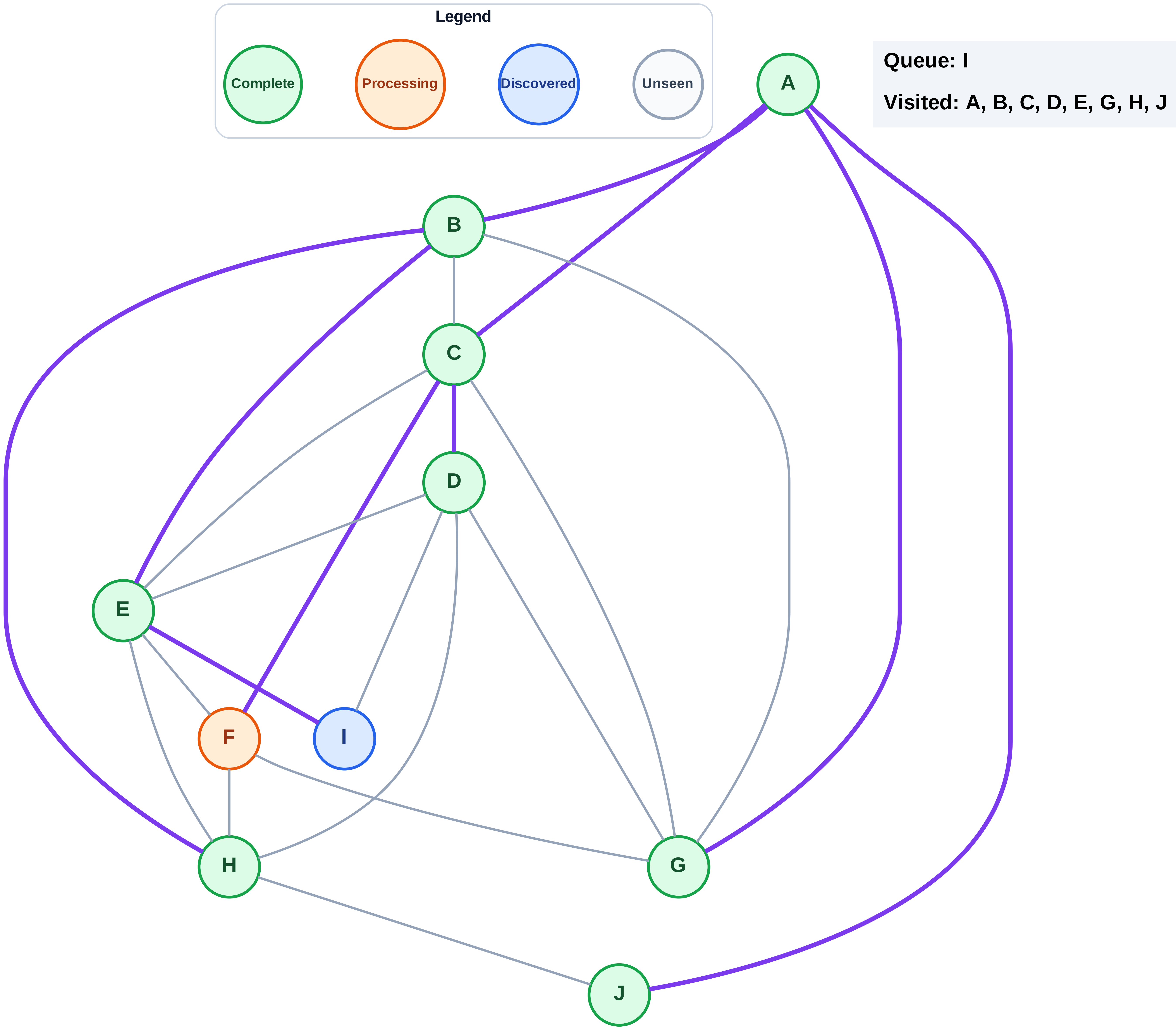
BFS Traversal

Step 17: No new neighbors from D



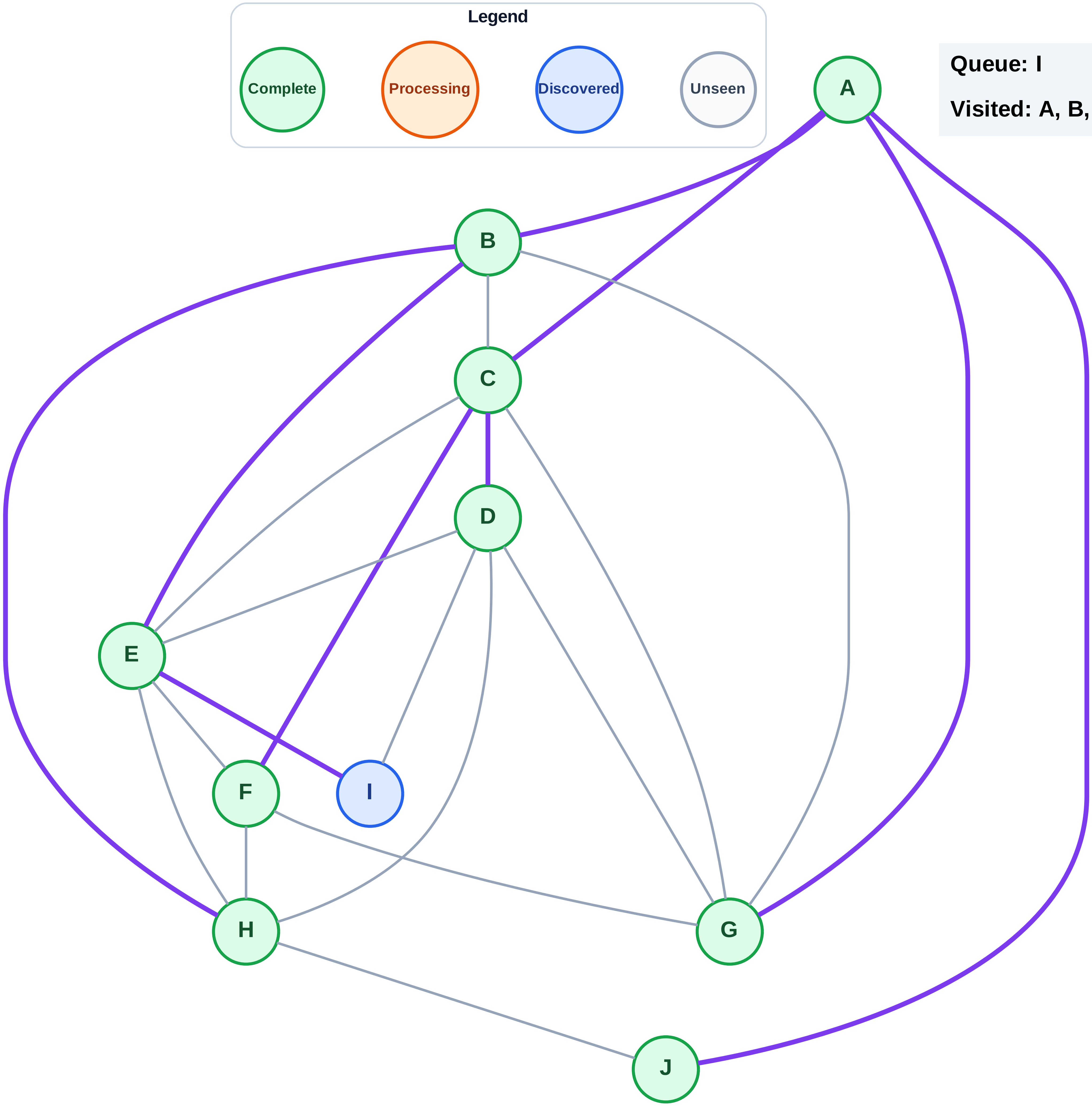
BFS Traversal

Step 18: Process node F



BFS Traversal

Step 19: No new neighbors from F

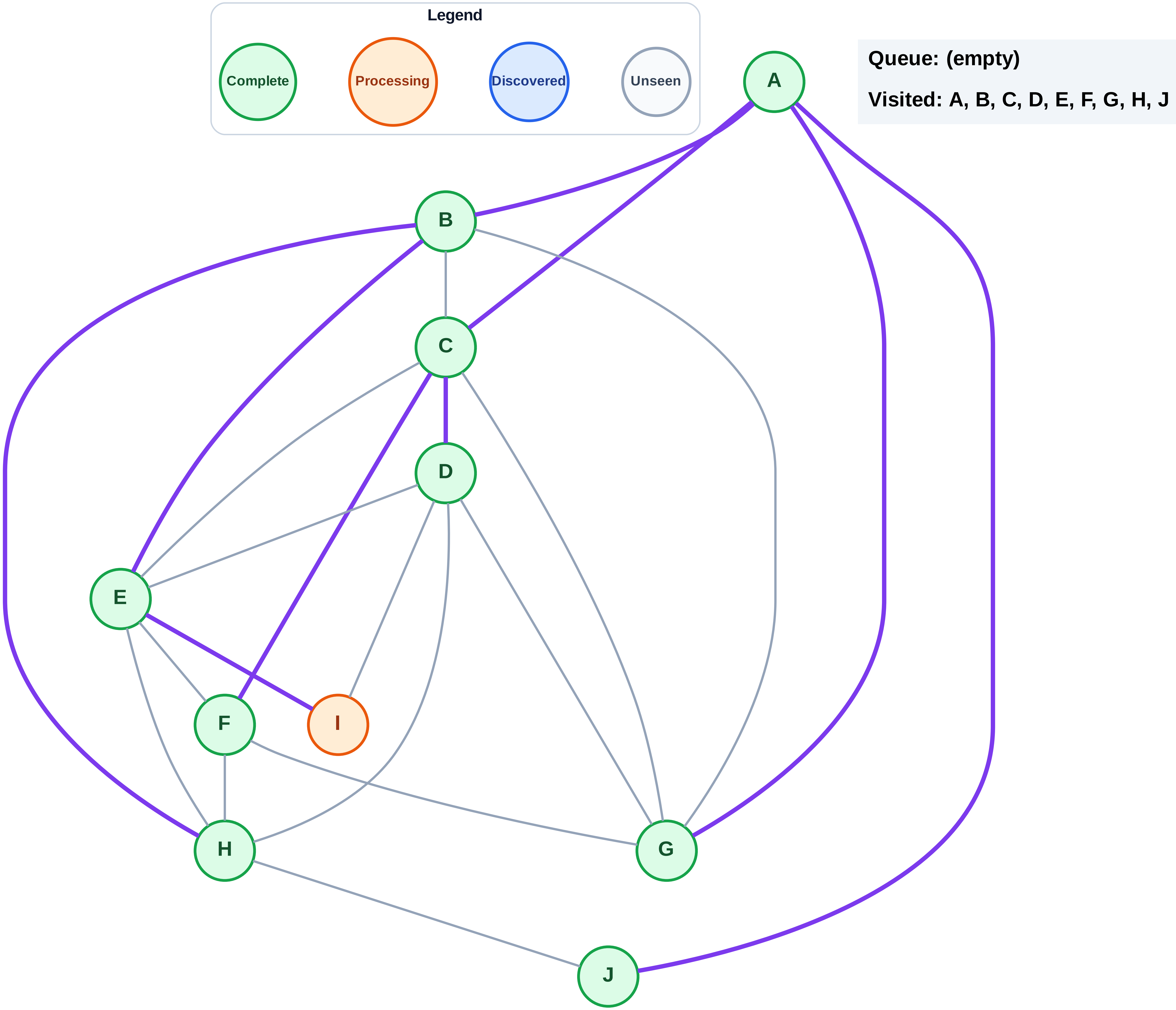


Queue: I

Visited: A, B, C, D, E, F, G, H, J

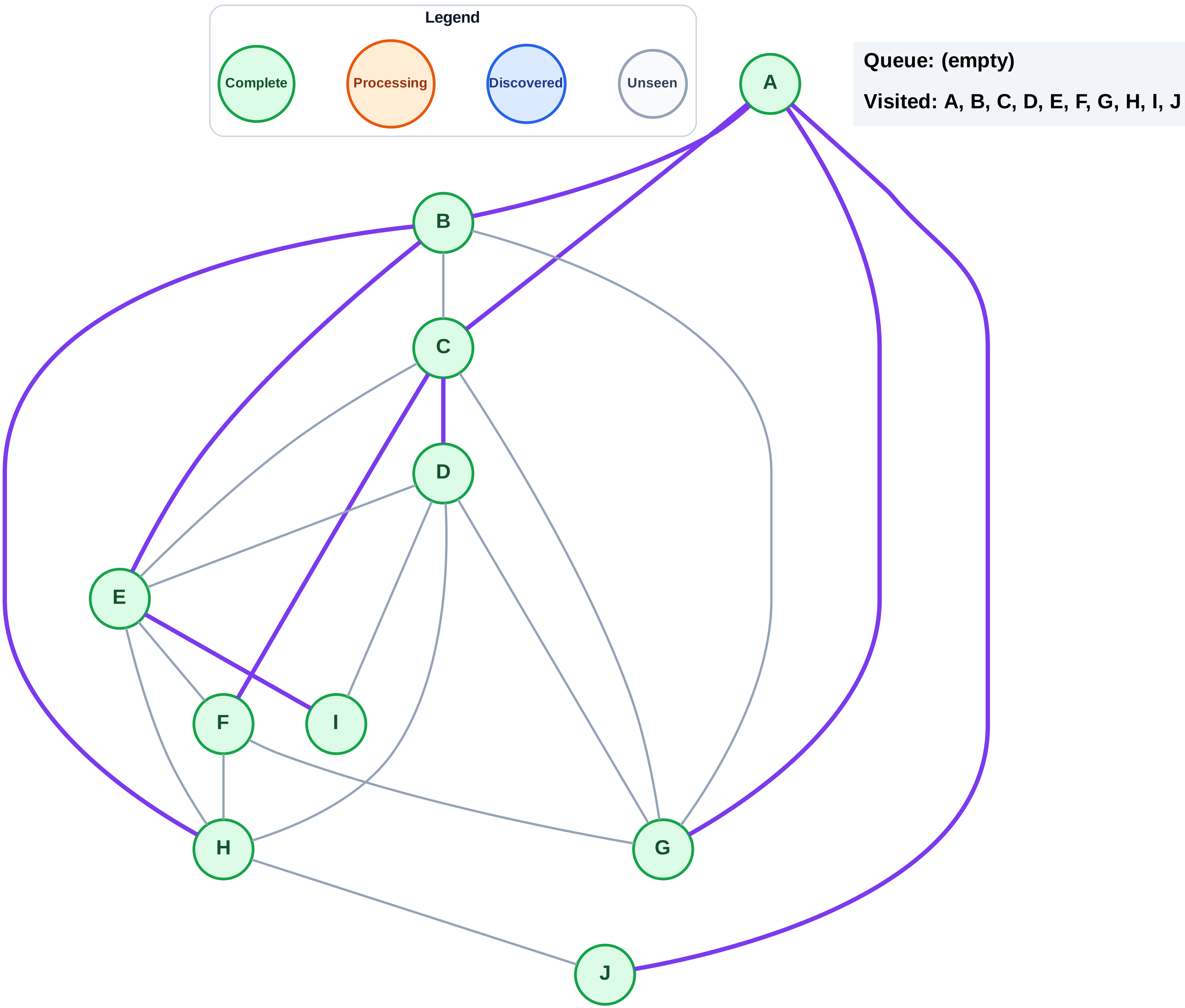
BFS Traversal

Step 20: Process node I



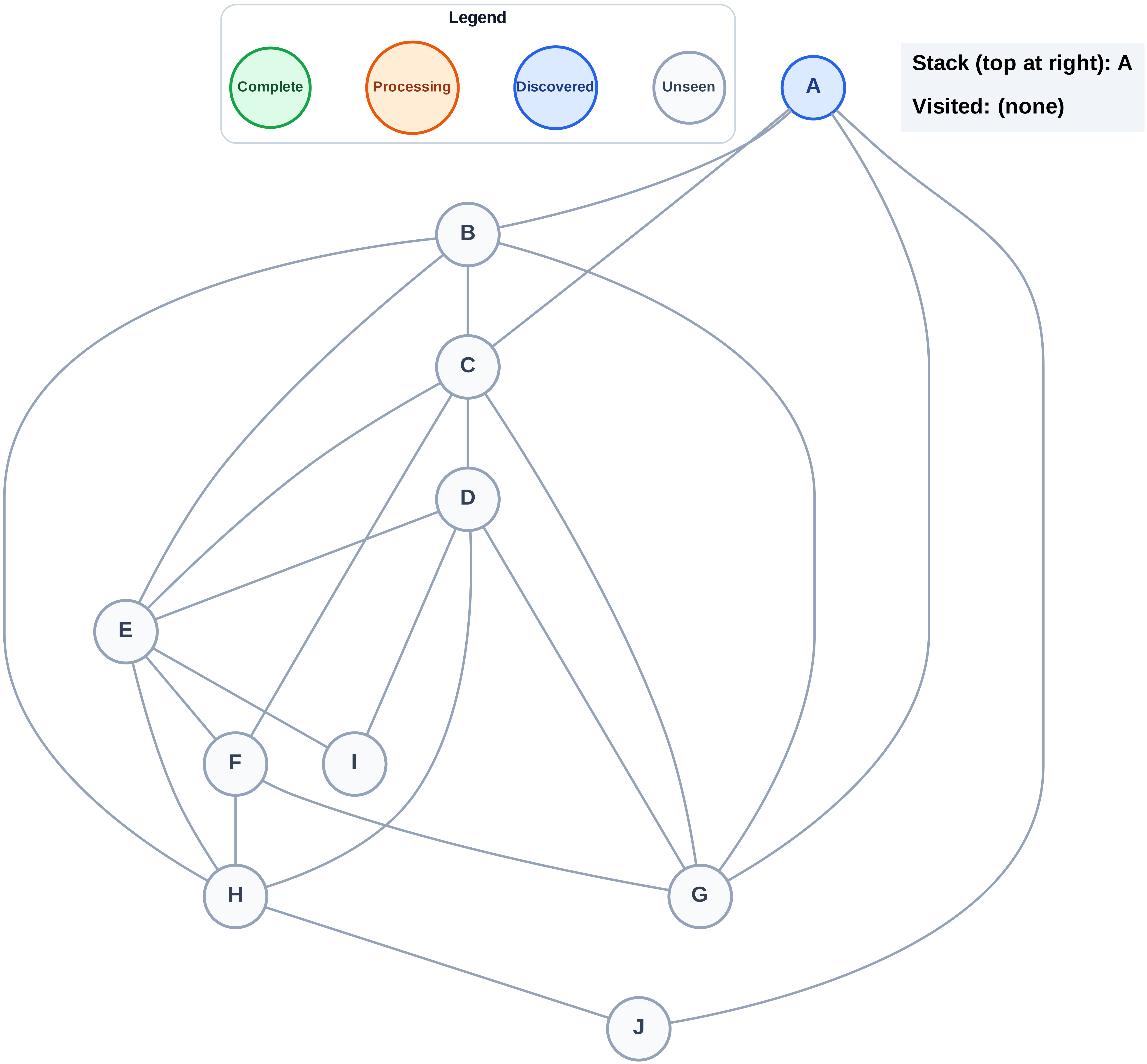
BFS Traversal

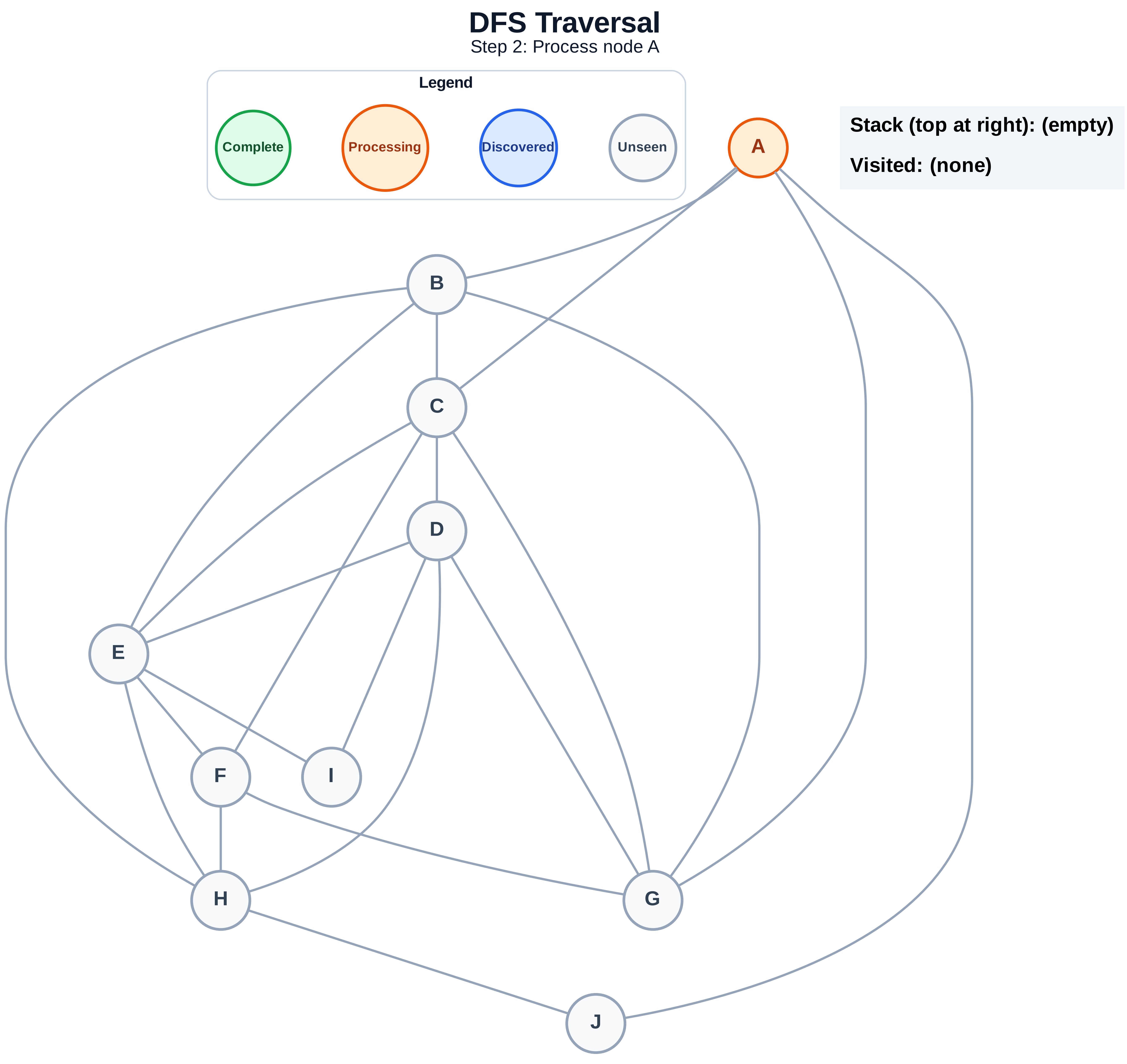
Step 21: No new neighbors from I



DFS Traversal

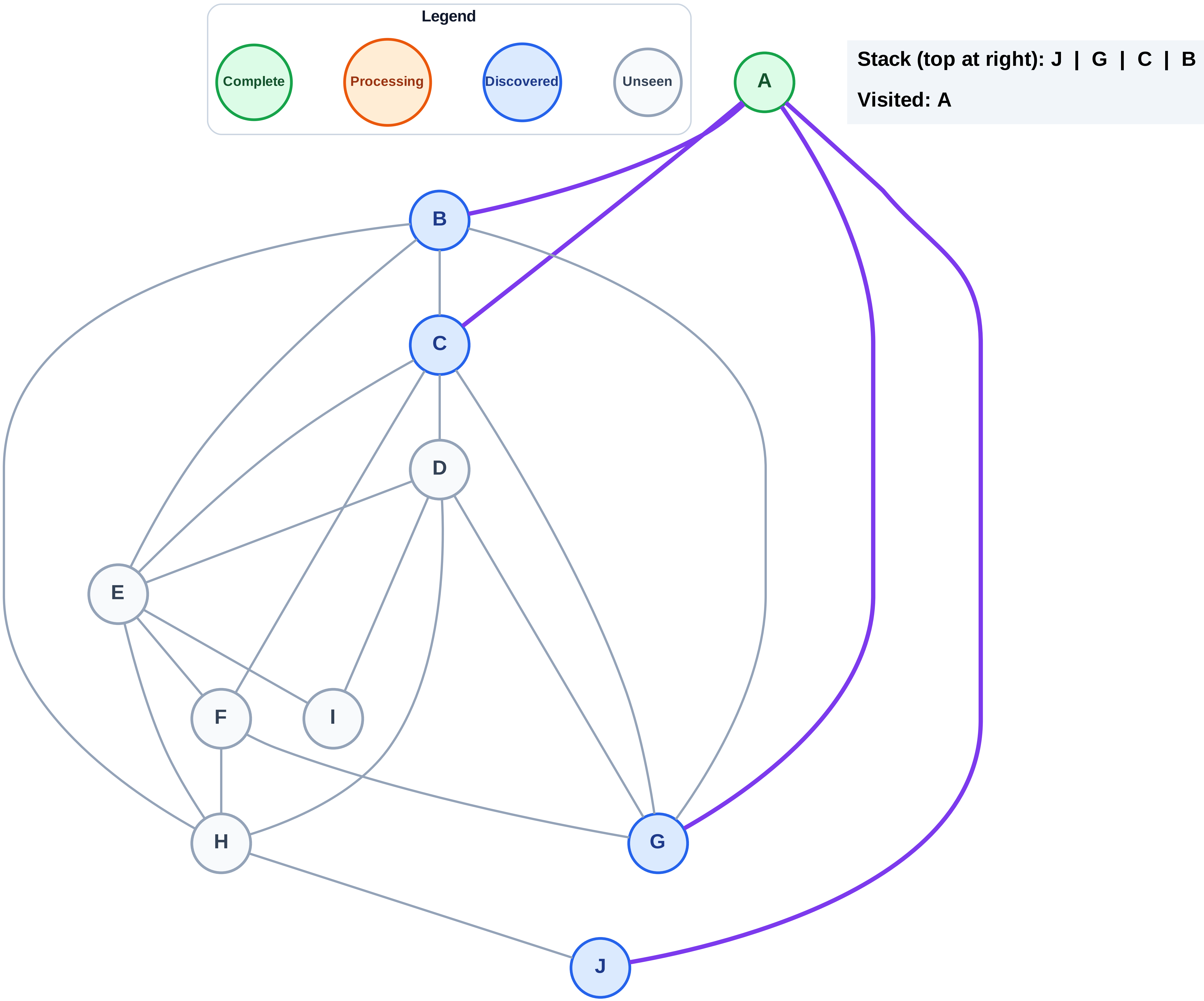
Step 1: Start at A




DFS Traversal
Step 2: Process node A
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): (empty)
Visited: (none)
A
B
C
D
E
F
I
H
G
J

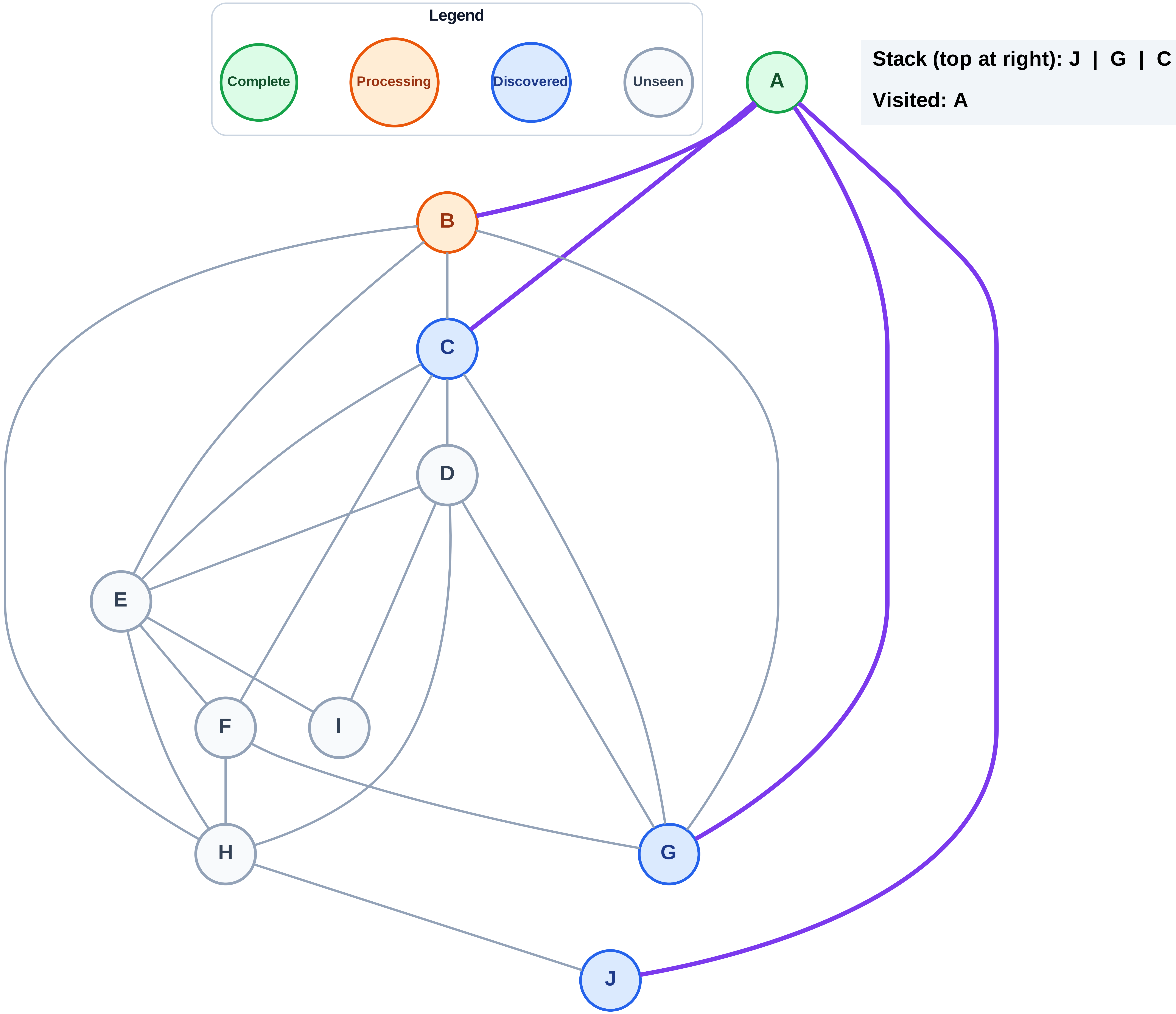
DFS Traversal

Step 3: Discover J, G, C, B from A



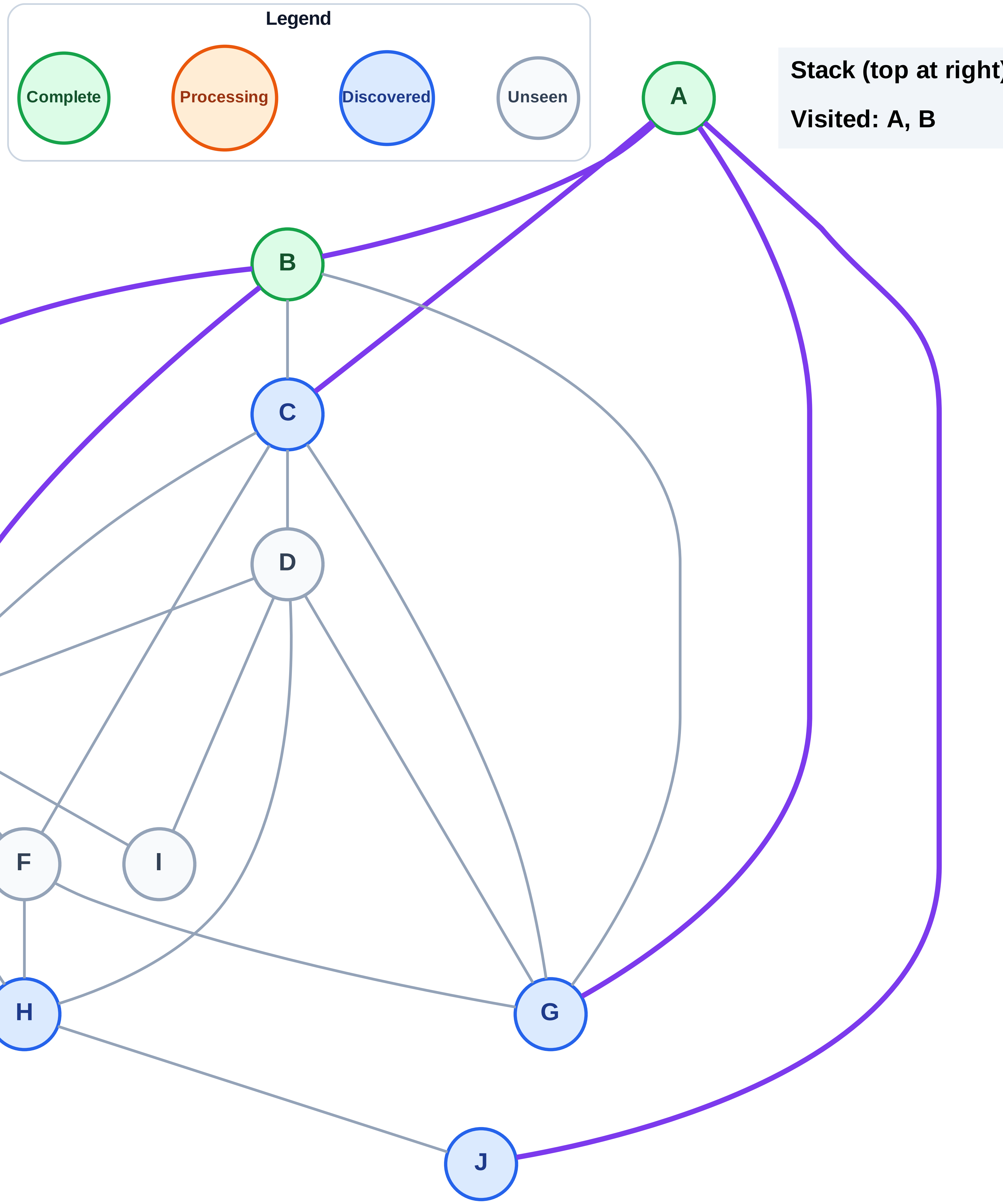
DFS Traversal

Step 4: Process node B



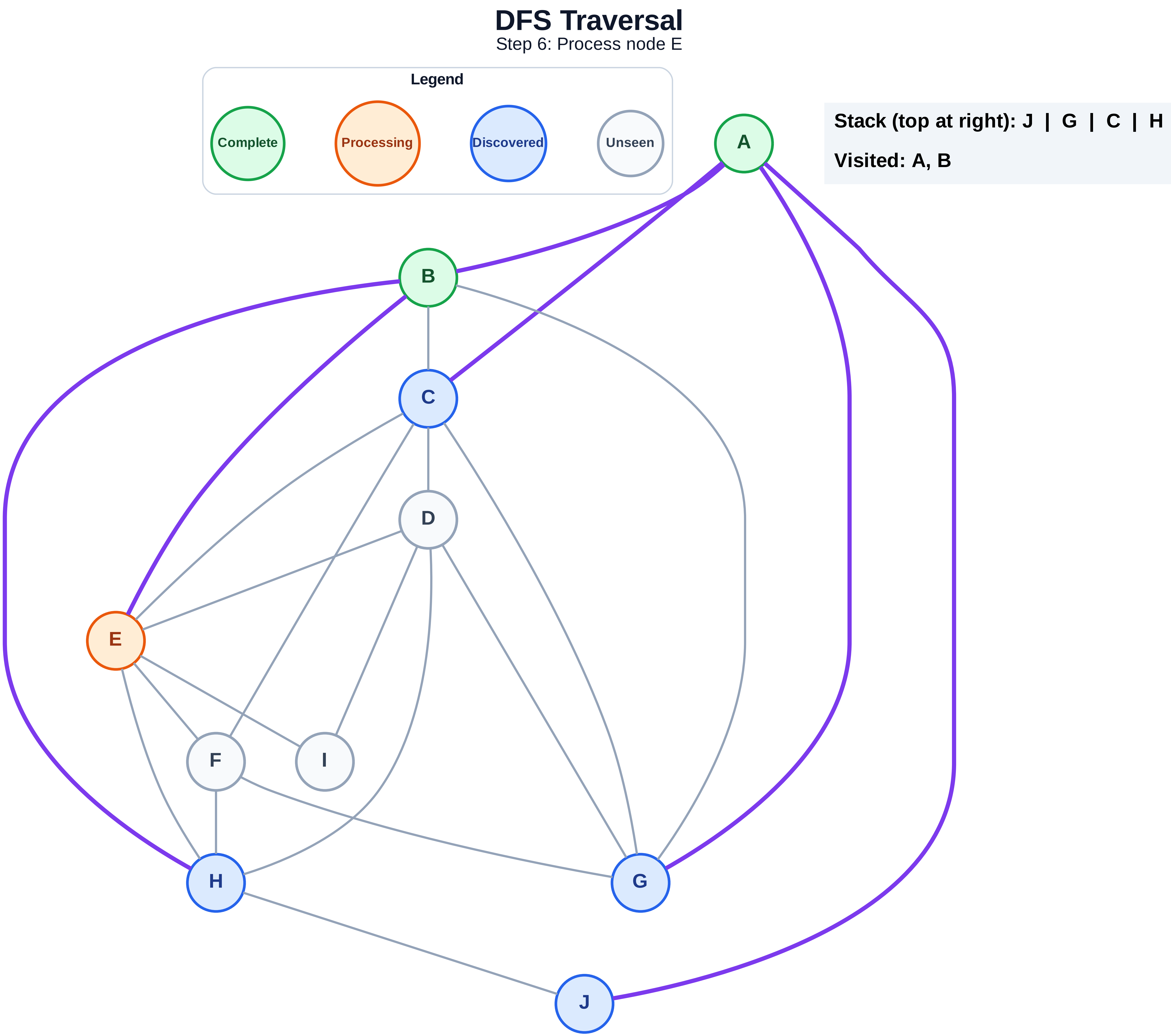
DFS Traversal

Step 5: Discover H, E from B



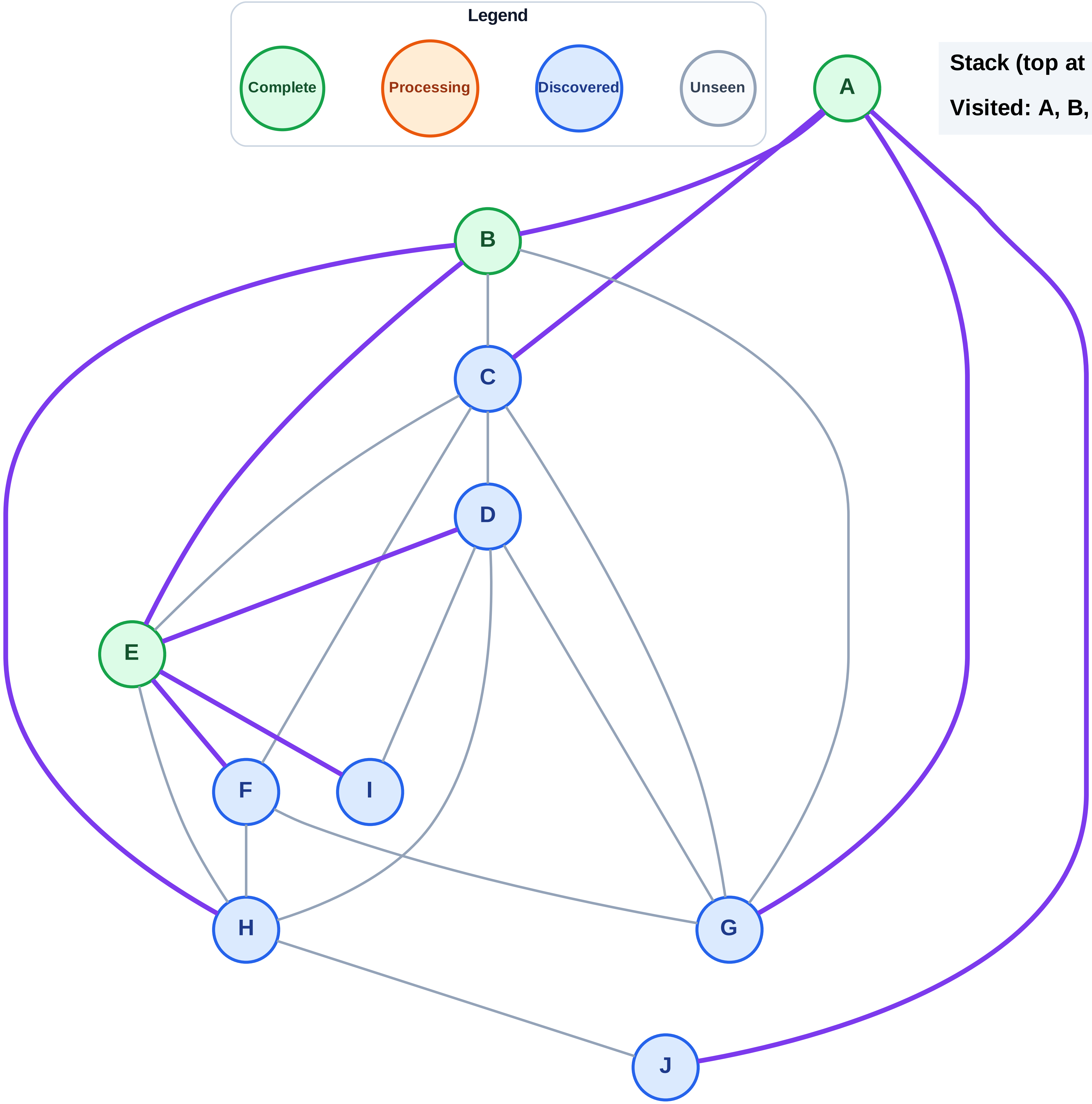
Stack (top at right): J | G | C | H | E

Visited: A, B


DFS Traversal
Step 6: Process node E
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): J | G | C | H
Visited: A, B
A
B
C
D
E
F
I
H
G
J

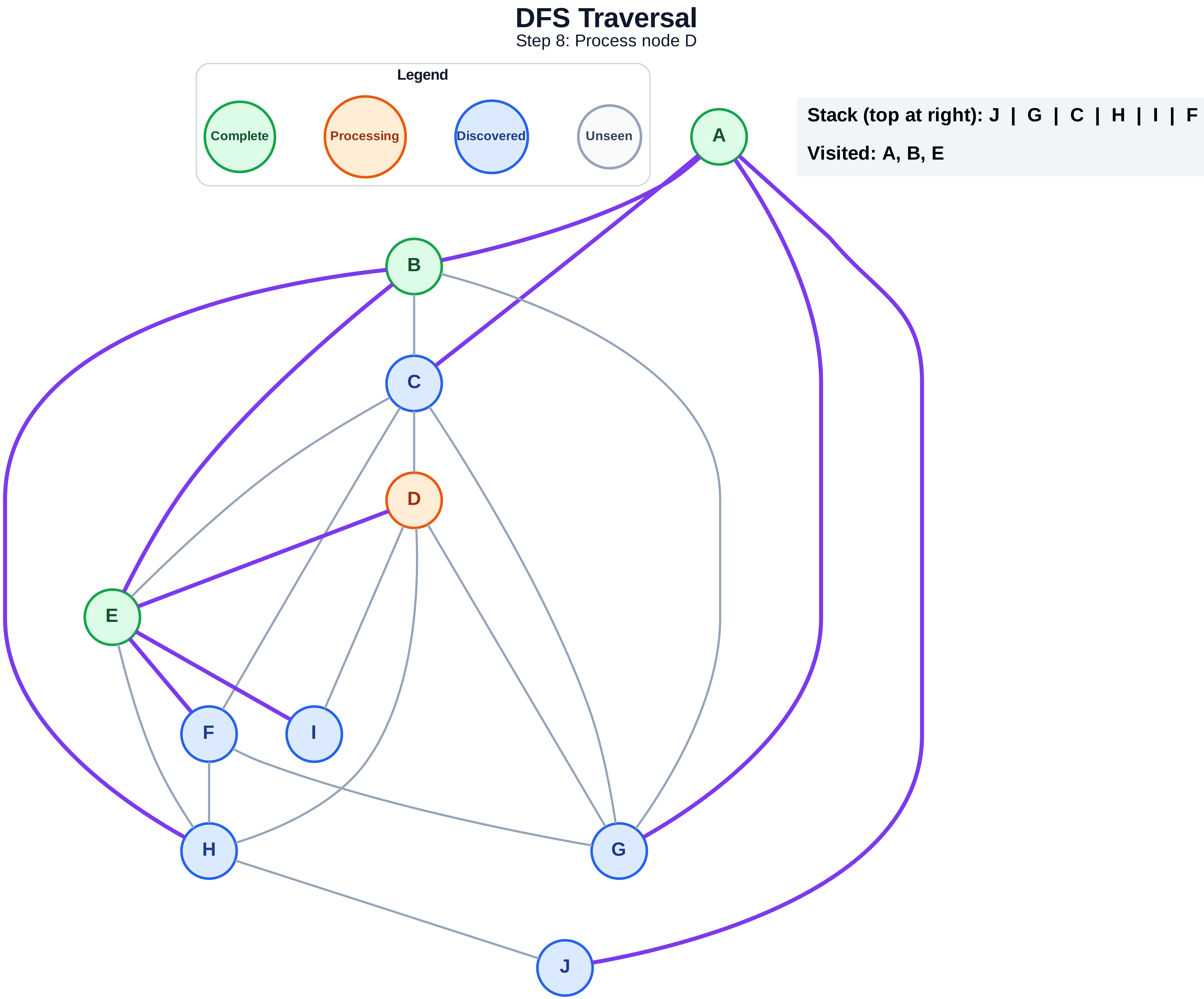
DFS Traversal

Step 7: Discover I, F, D from E



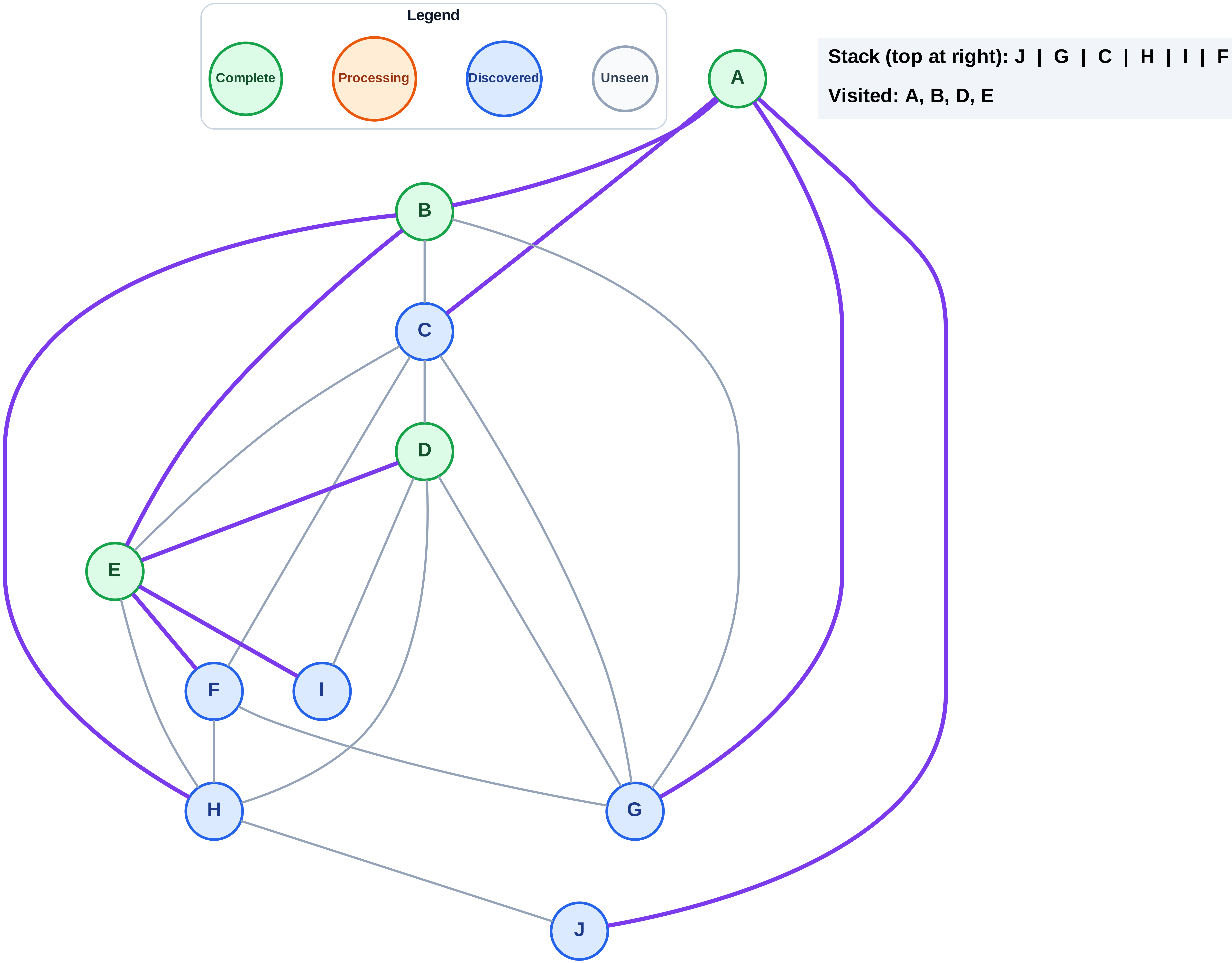
Stack (top at right): J | G | C | H | I | F | D

Visited: A, B, E


DFS Traversal
Step 8: Process node D
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): J | G | C | H | I | F
Visited: A, B, E
A
B
C
D
E
F
I
H
G
J

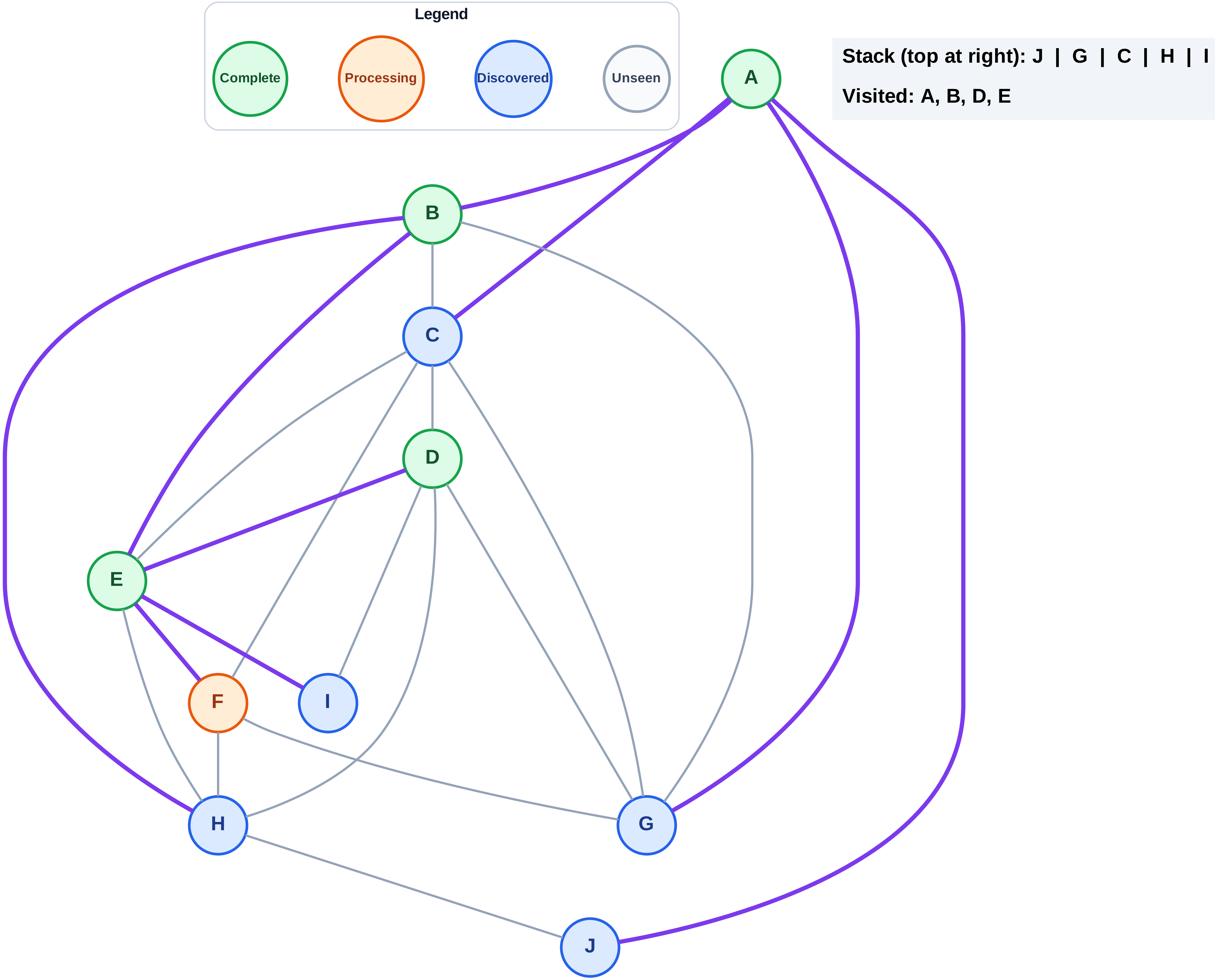
DFS Traversal

Step 9: No new neighbors from D



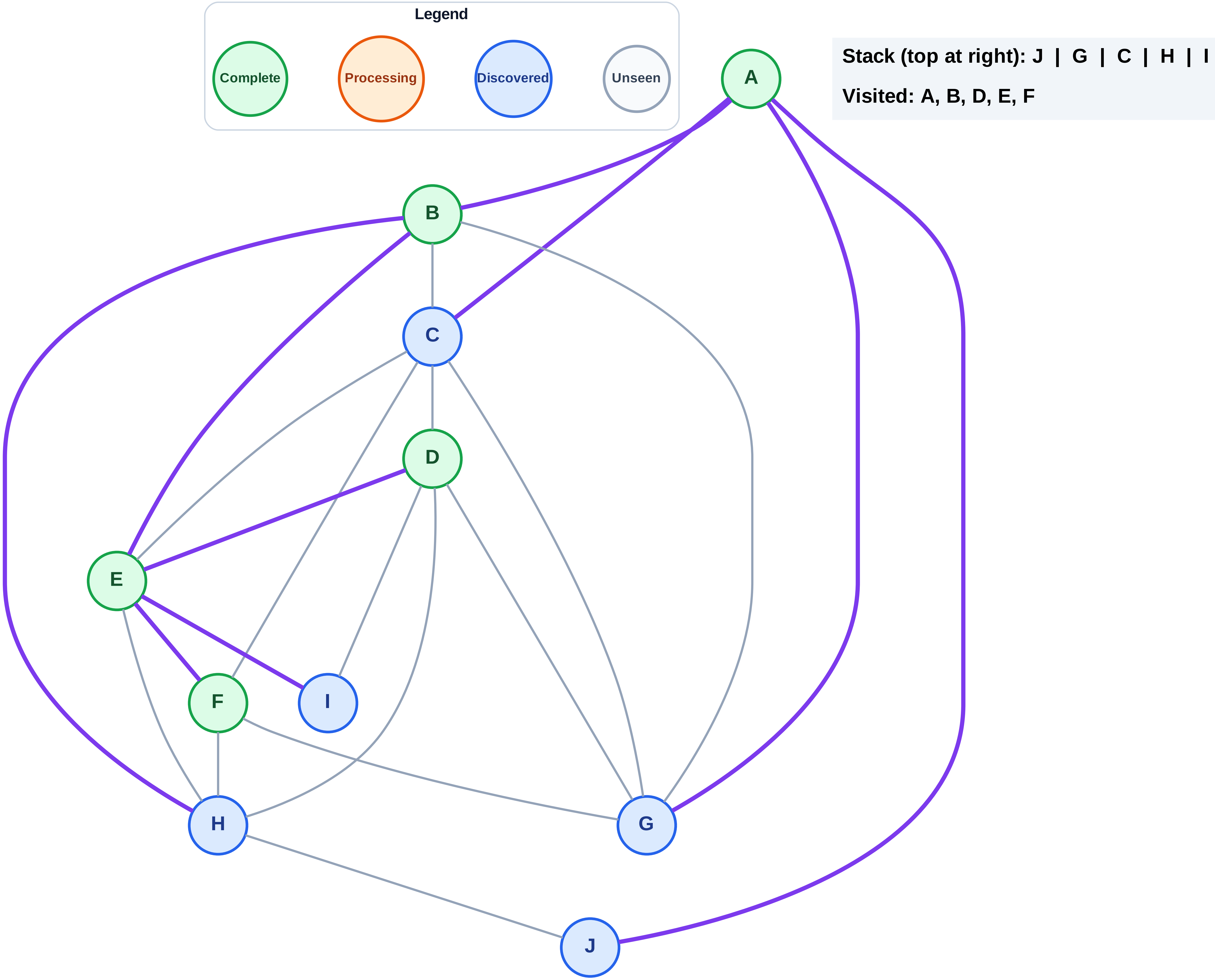
DFS Traversal

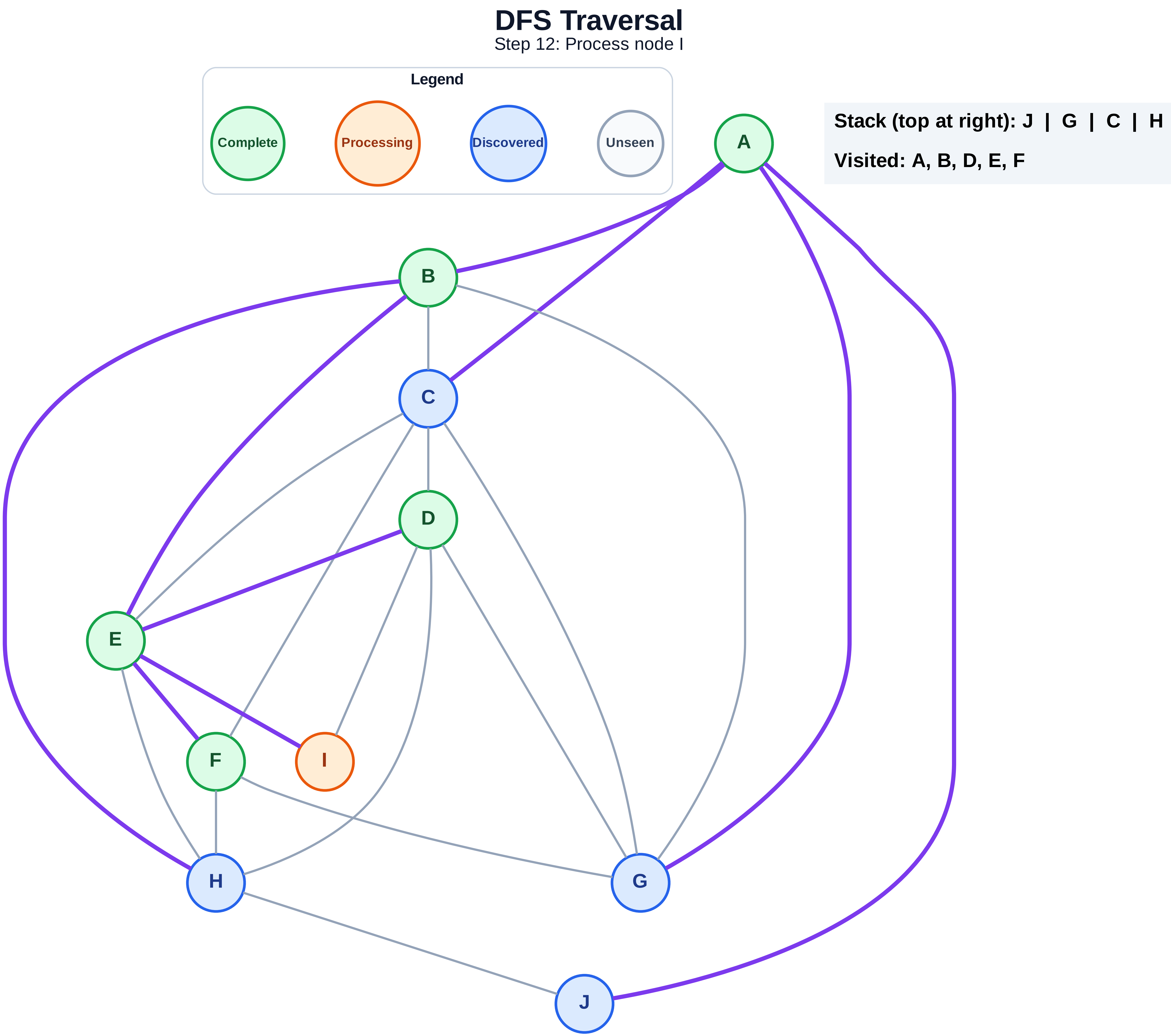
Step 10: Process node F



DFS Traversal

Step 11: No new neighbors from F

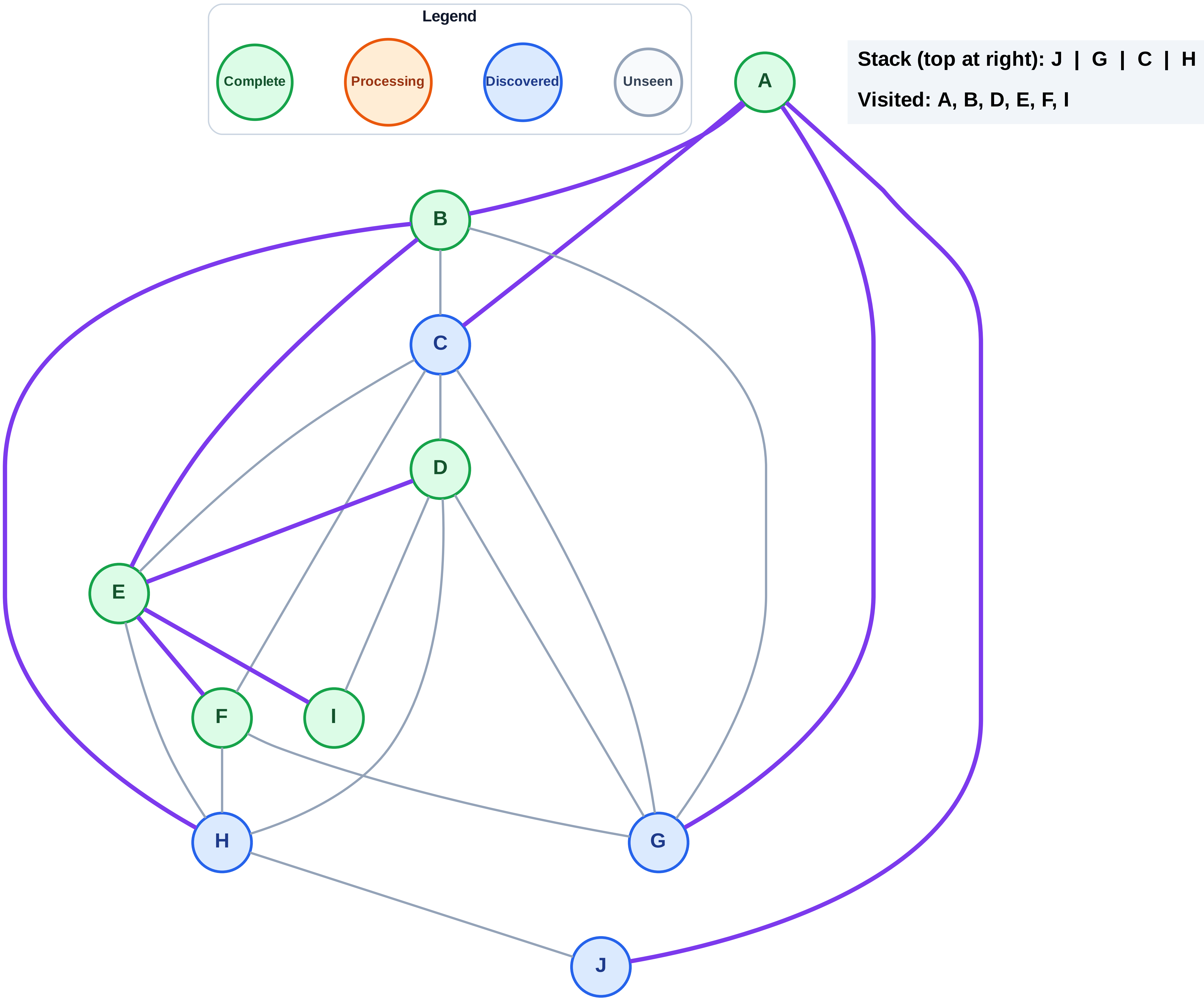




DFS Traversal
Step 12: Process node I
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): J | G | C | H
Visited: A, B, D, E, F
A
B
C
D
E
F
I
H
G
J

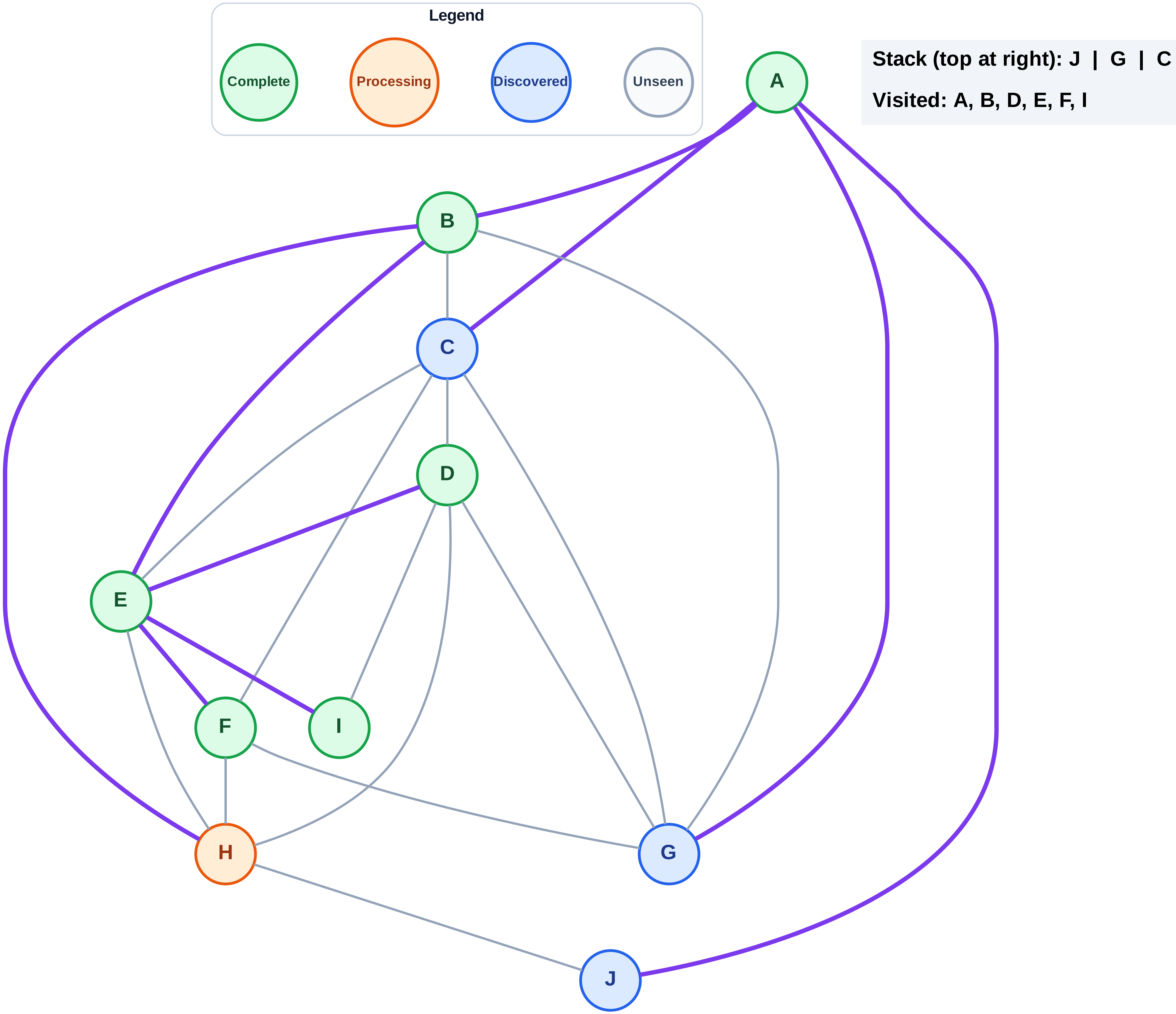
DFS Traversal

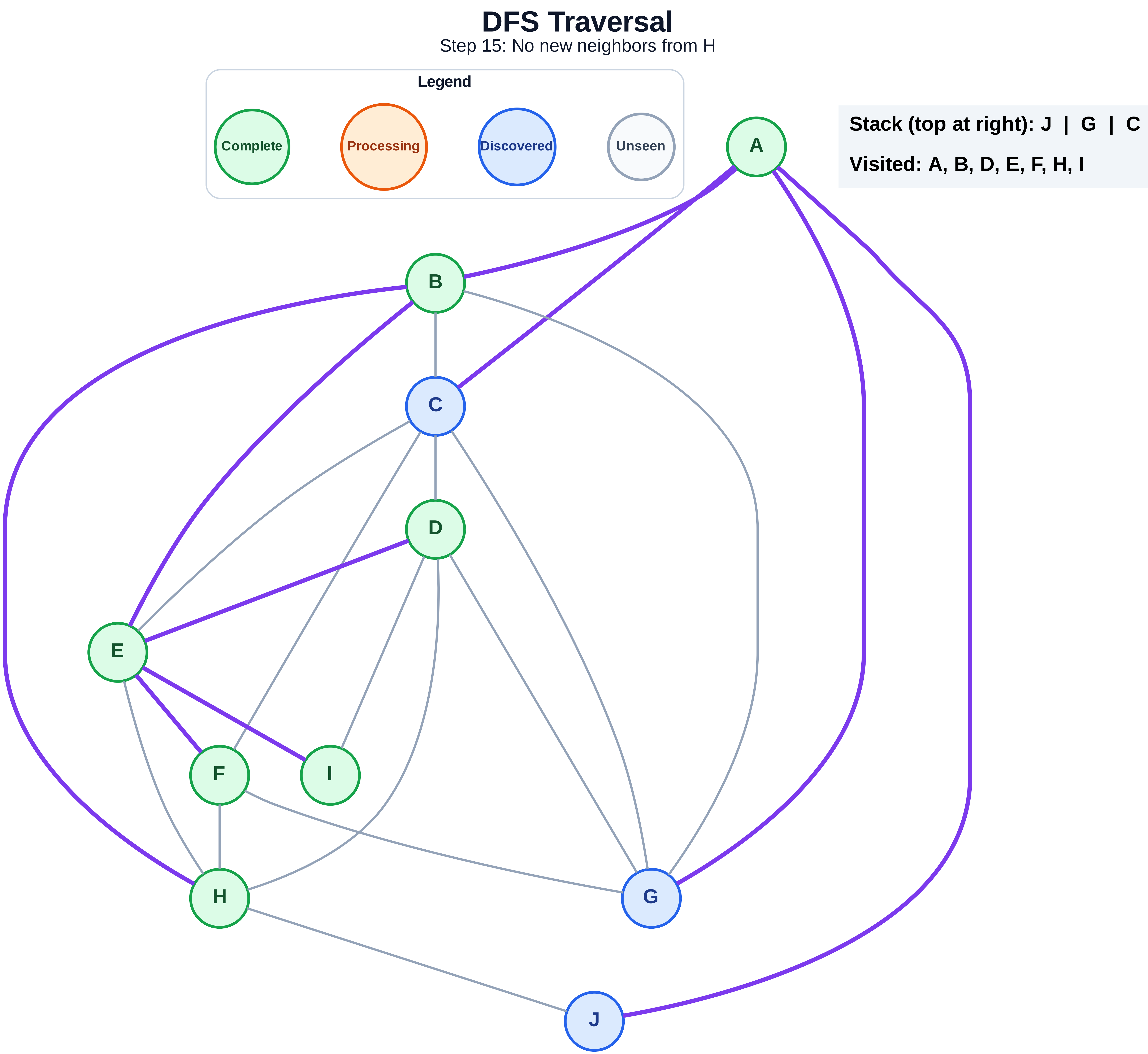
Step 13: No new neighbors from I



DFS Traversal

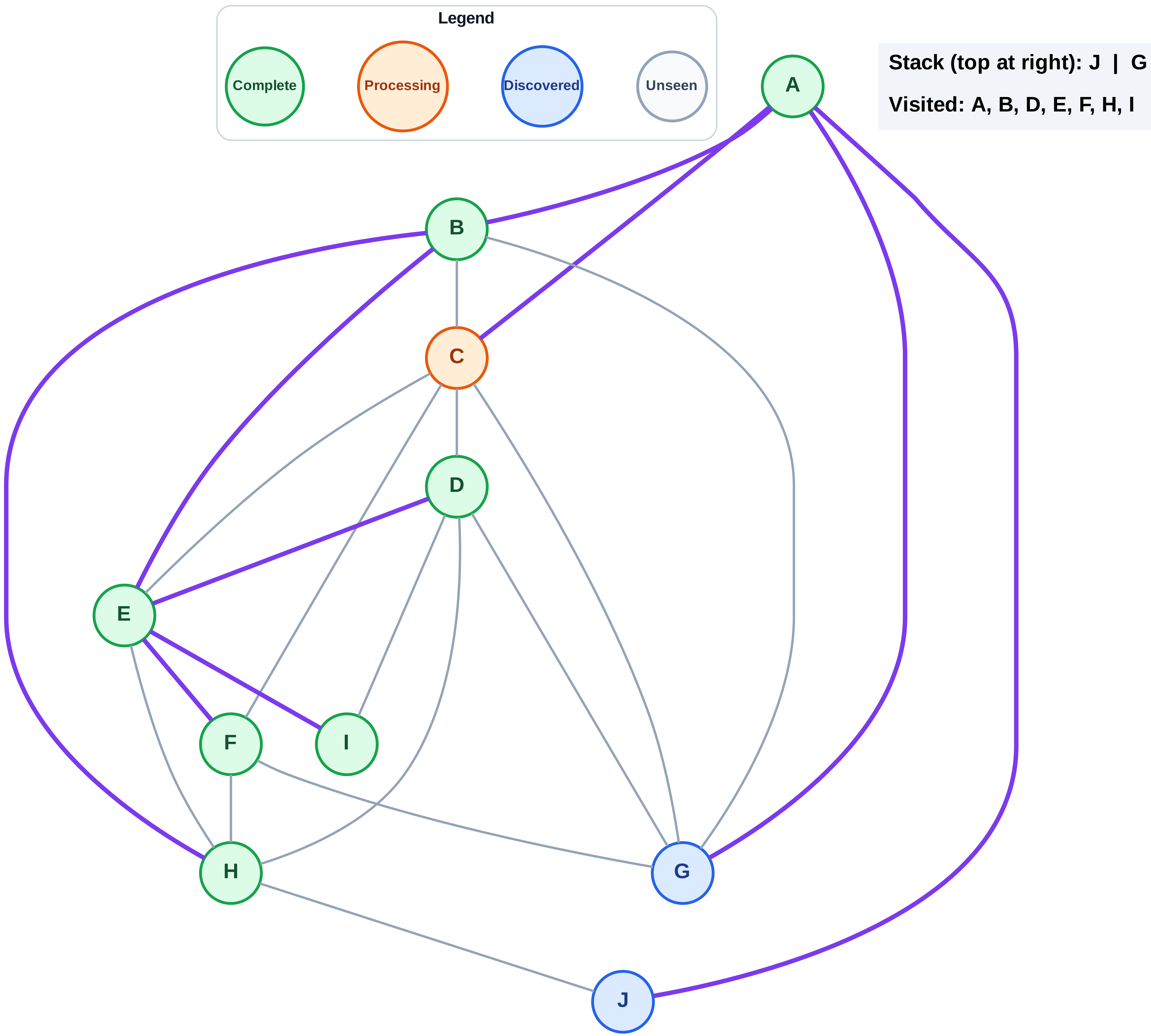
Step 14: Process node H




DFS Traversal
Step 15: No new neighbors from H
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): J | G | C
Visited: A, B, D, E, F, H, I
A
B
C
D
E
F
I
H
G
J

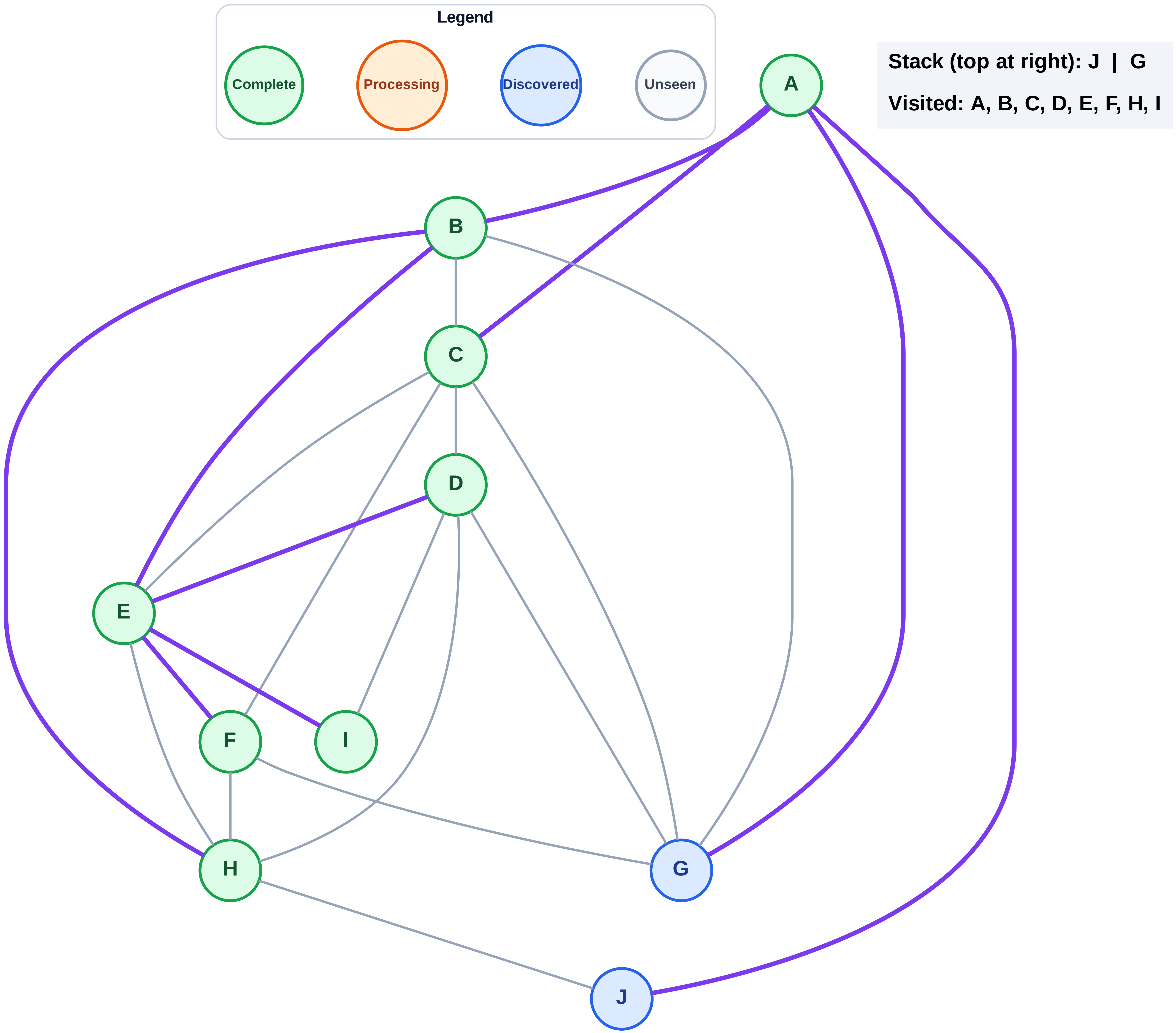
DFS Traversal

Step 16: Process node C



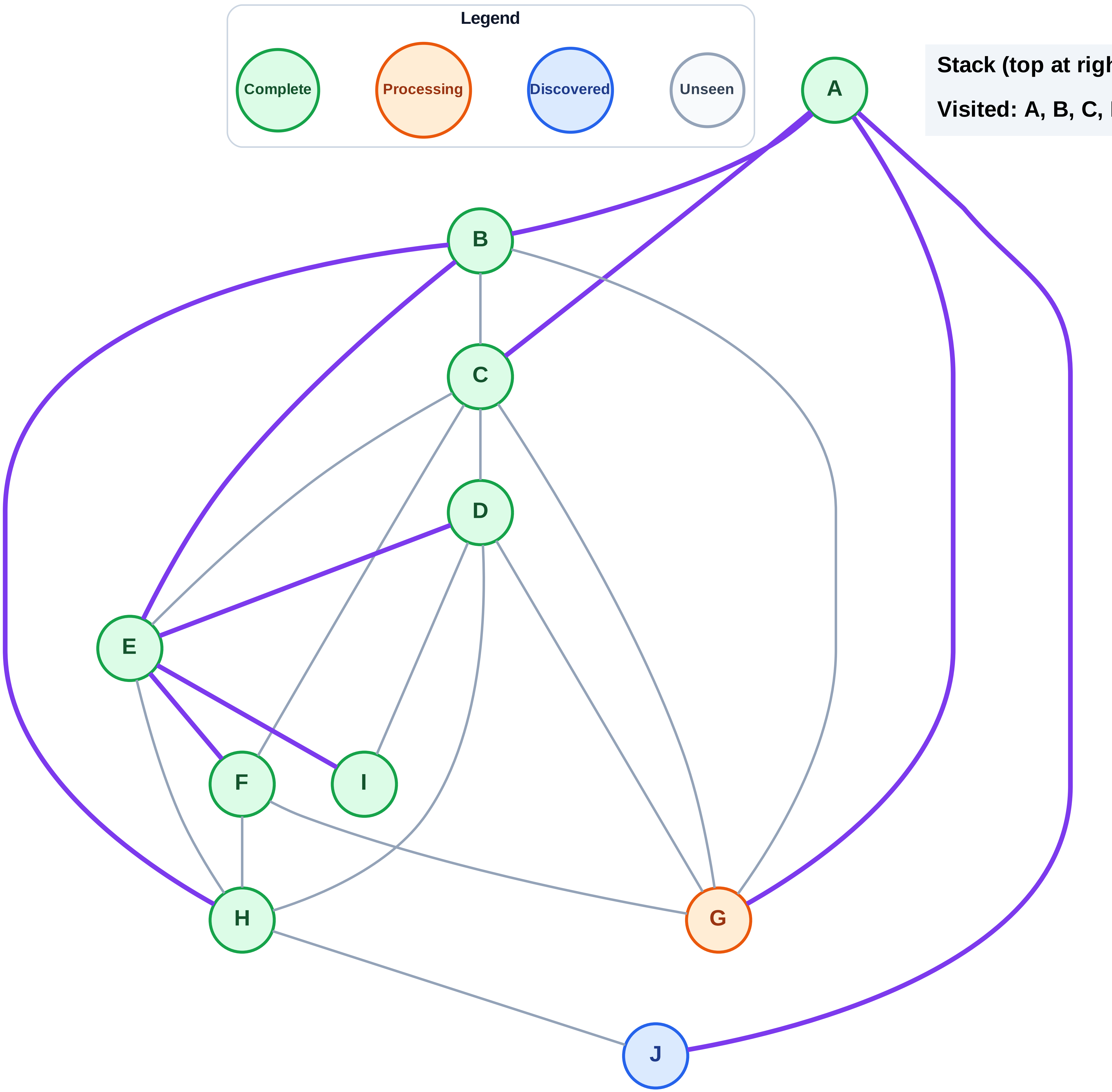
DFS Traversal

Step 17: No new neighbors from C



DFS Traversal

Step 18: Process node G

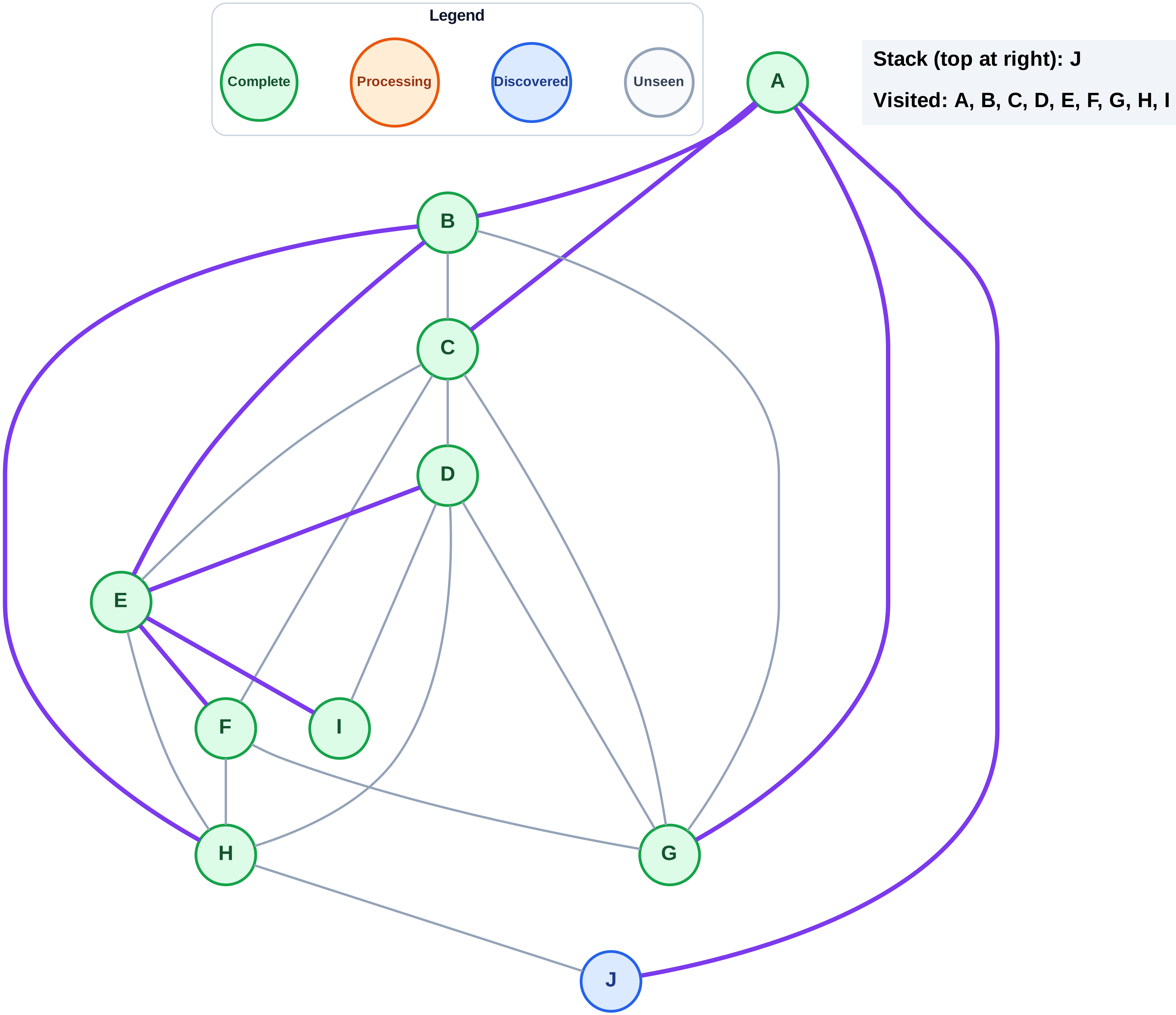


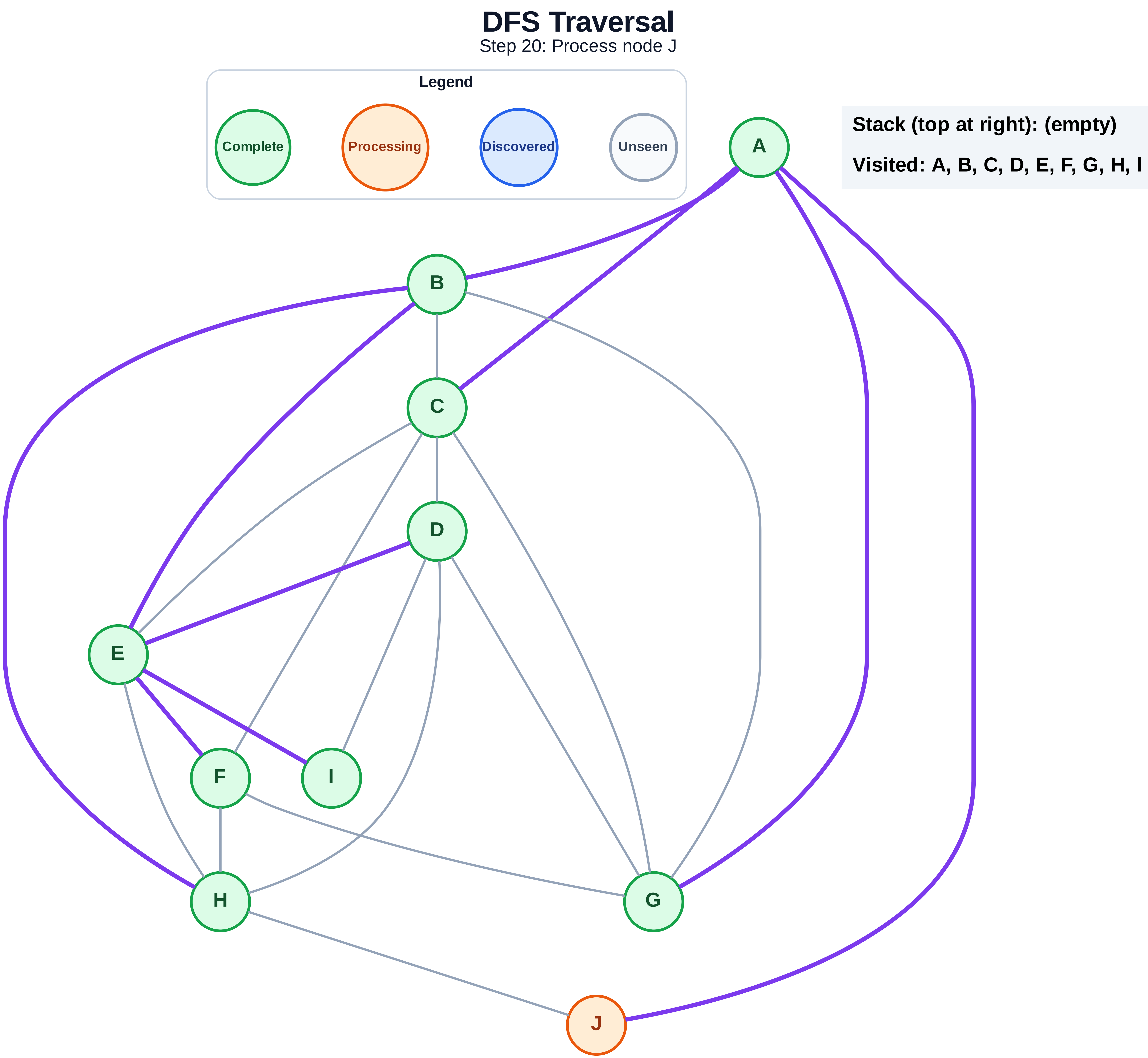
Stack (top at right): J

Visited: A, B, C, D, E, F, H, I

DFS Traversal

Step 19: No new neighbors from G




DFS Traversal
Step 20: Process node J
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I
A
B
C
D
E
F
I
H
G
J

DFS Traversal

Step 21: No new neighbors from J